

LMM Report – HGS_MAX

Generated: 2026-07-08 14:19

Imputations: 10

Random slope: FALSE | Interaction: FALSE

Exposures: 1 | Covariate sets: 2

Model Configuration

Outcome column: HGS_MAX
Outcome transform: identity
Estimation method: REML = FALSE (Maximum Likelihood)
Random slope: FALSE
Interaction (exposure × time): FALSE

Exposures:

	exposure	exposure_type	ref_level
1	prot_content_dairy_100g	linear	NA

Covariate sets:

main:
age_at_baseline_scaled, BMI_category, education_level, smoking_status, pa_levels_tertile_f1, diabetes_status, sumtot1_scaled

other_PA:
age_at_baseline_scaled, BMI_category, education_level, smoking_status, pa_levels_who_f1, diabetes_status, sumtot1_scaled

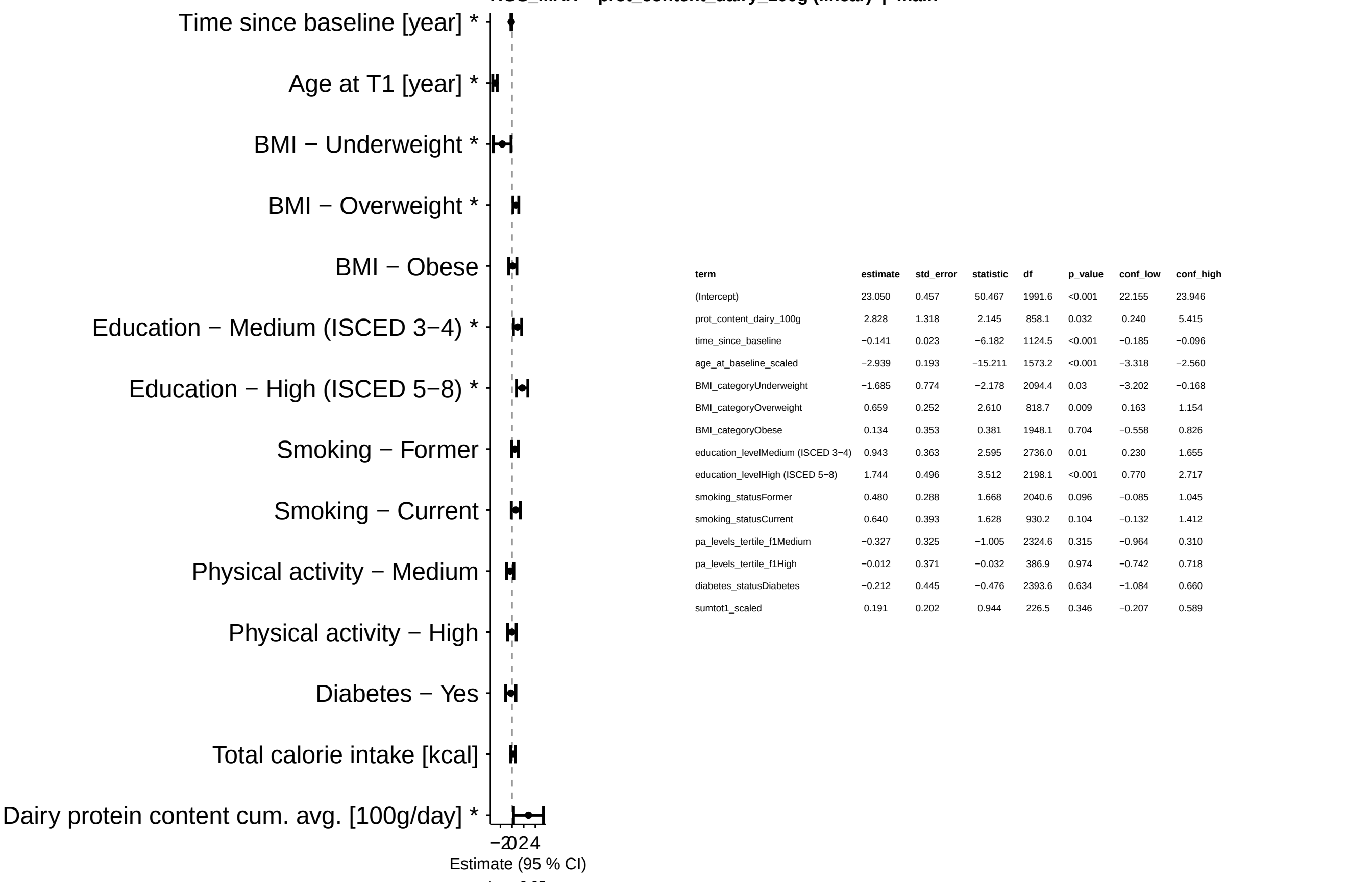
Covariate set: main

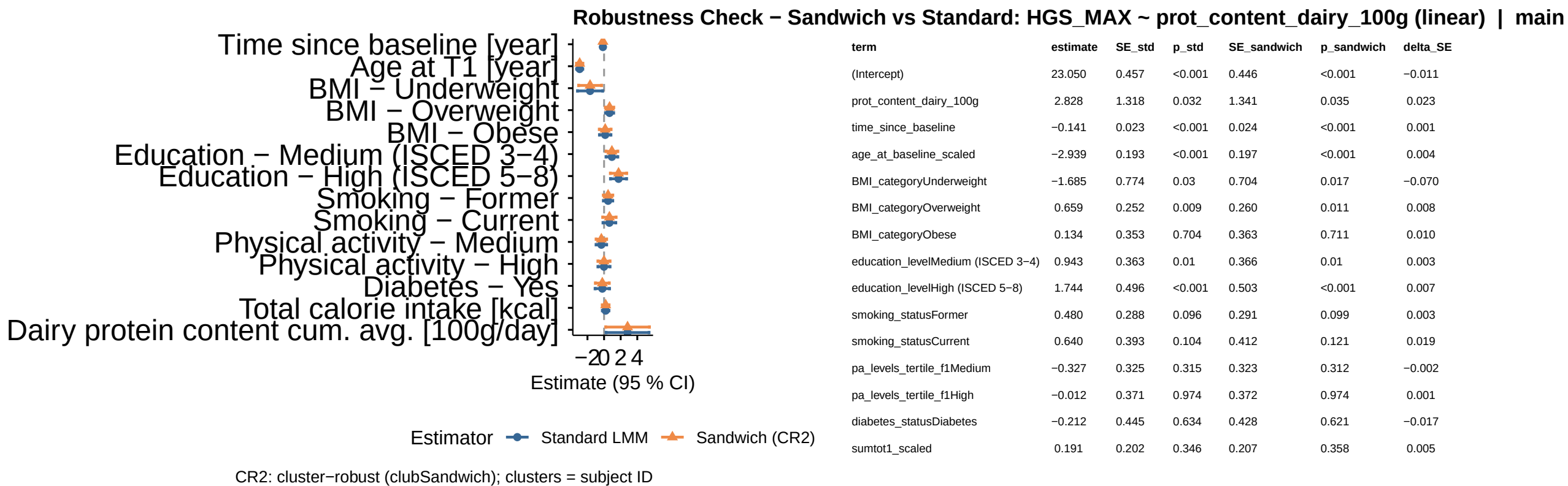
age_at_baseline_scaled · BMI_category · education_level · smoking_status · pa_levels_tertile_f1 · diabetes_status · sumtot1_scaled

Formula – HGS_MAX ~ prot_content_dairy_100g (linear) | main

HGS_MAX ~ prot_content_dairy_100g + time_since_baseline + age_at_baseline_scaled +
BMI_category + education_level + smoking_status + pa_levels_tertile_f1 +
diabetes_status + sumtot1_scaled + (1 | pt)

Imputations pooled: 10





Variance Components – HGS_MAX ~ prot_content_dairy_100g (linear) | main

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.. Full VarCorr output .....
Groups   Name          Variance Std.Dev.
pt       (Intercept) 14.808    3.8482
Residual              14.340    3.7868

.. Summary .....
Random intercept variance: 14.8084 (SD = 3.84817)
Random slope: not included in this model
Residual variance:         14.34015 (SD = 3.78684)
```

Effect Size (R²) – HGS_MAX ~ prot_content_dairy_100g (linear) | main

Marginal R² (fixed effects only): 0.1616

Conditional R² (fixed + random effects): 0.5875

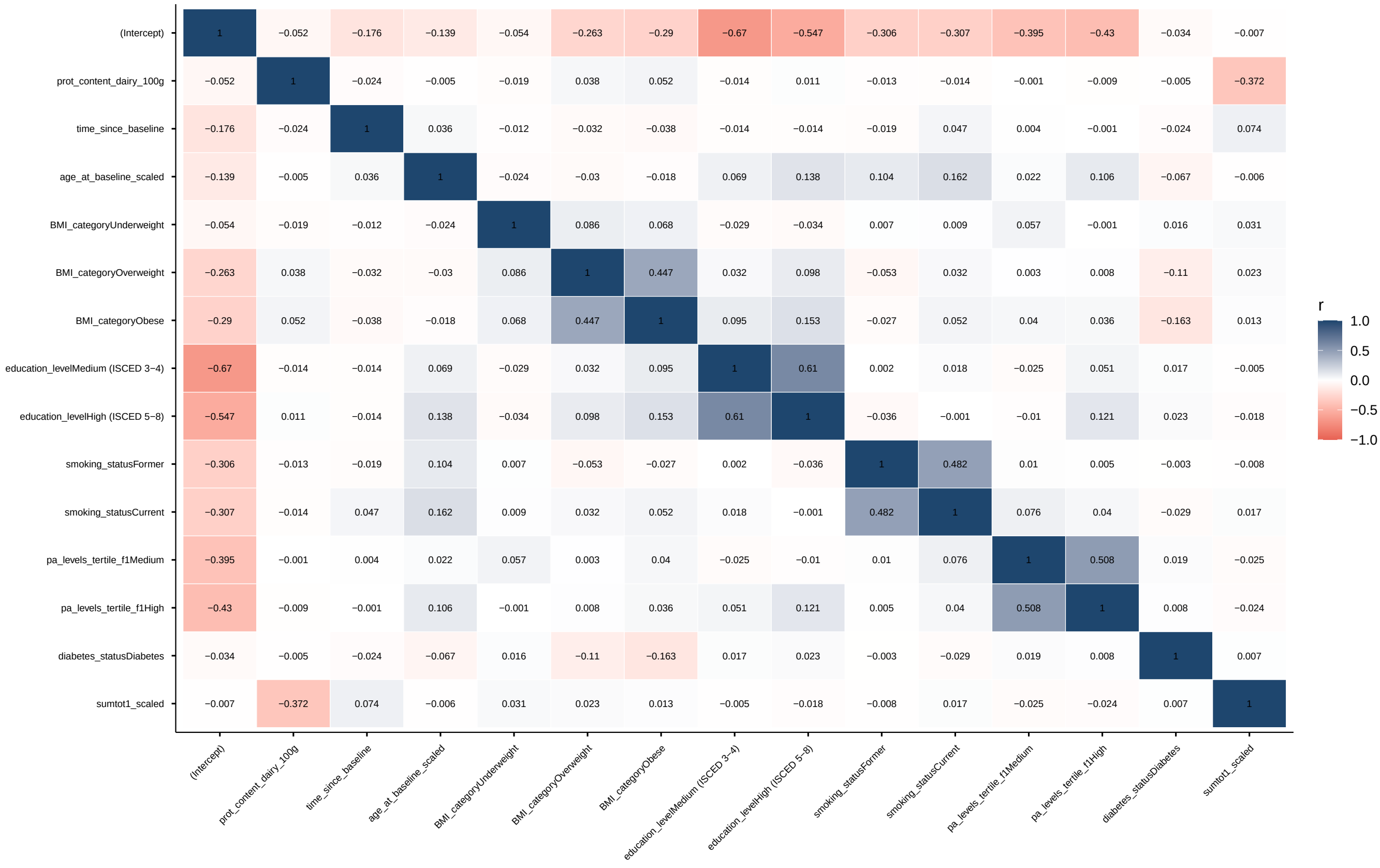
Method: Nakagawa & Schielzeth (2013), Johnson (2014)

VIF – HGS_MAX ~ prot_content_dairy_100g (linear) | main

VIF > 5 flagged with '!'

	term	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$	adjusted_vif	flag
	prot_content_dairy_100g	1.167605	1	1.080558	1.081	
	time_since_baseline	1.014461	1	1.007205	1.007	
	age_at_baseline_scaled	1.069430	1	1.034133	1.034	
	BMI_category	1.092713	3	1.014887	1.015	
	education_level	1.079269	2	1.019254	1.019	
	smoking_status	1.054191	2	1.013281	1.013	
	pa_levels_tertile_f1	1.046996	2	1.011547	1.012	
	diabetes_status	1.041343	1	1.020462	1.020	
	sumtot1_scaled	1.172093	1	1.082633	1.083	

Fixed-Effect Correlations – HGS_MAX ~ prot_content_dairy_100g (linear) | main



Influential Points – HGS_MAX ~ prot_content_dairy_100g (linear) | main

Influential observations (top 1% Cook's D > 0.20117): 32

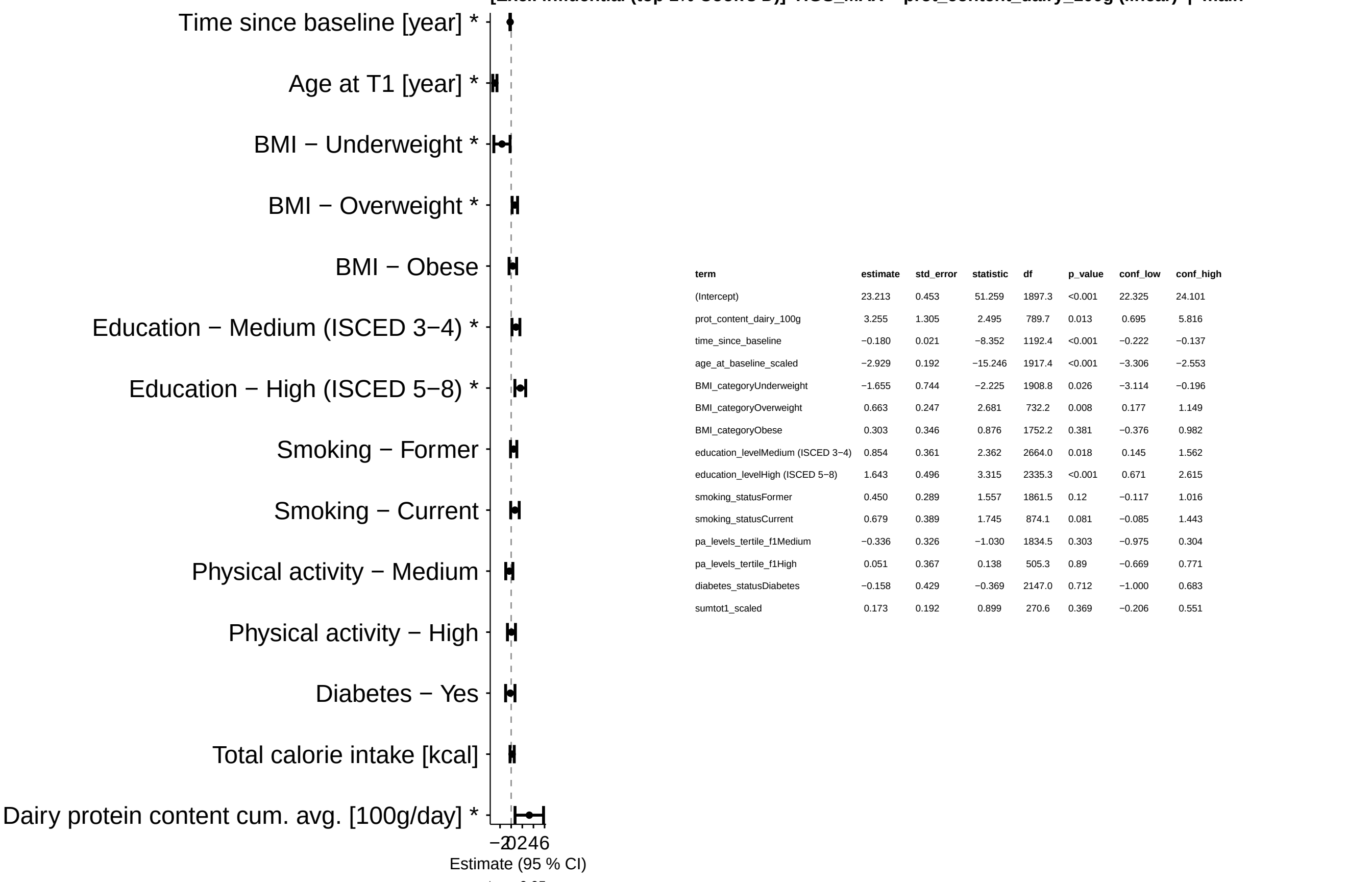
Index	Cook's D
index	cooks_d
22	0.28608
516	0.25491
837	0.36168
896	0.29306
1227	0.29382
1252	0.43923
1428	0.21063
1758	0.23953
1782	0.20816
1900	0.22865
1978	0.30997
1993	0.22428
2051	0.22533
2076	0.20127
2091	0.21184
2100	0.29933
2184	0.23058
2297	0.33113
2319	0.23290
2323	0.21514
2334	0.28472
2353	0.39466
2399	0.34287
2457	0.22062
2465	0.20298
2570	0.20762
2692	0.34566
2752	0.23739
2753	0.20561
2759	0.30189
2785	0.25793
2892	0.20140

Formula – [Excl. influential (top 1% Cook's D)] HGS_MAX ~ prot_content_dairy_100g (linear) | main

Excluded participants (top 1% Cook's D, imp 1): 31

Remaining n: 1104

HGS_MAX ~ prot_content_dairy_100g + time_since_baseline + age_at_baseline_scaled +
BMI_category + education_level + smoking_status + pa_levels_tertile_f1 +
diabetes_status + sumtot1_scaled + (1 | pt)



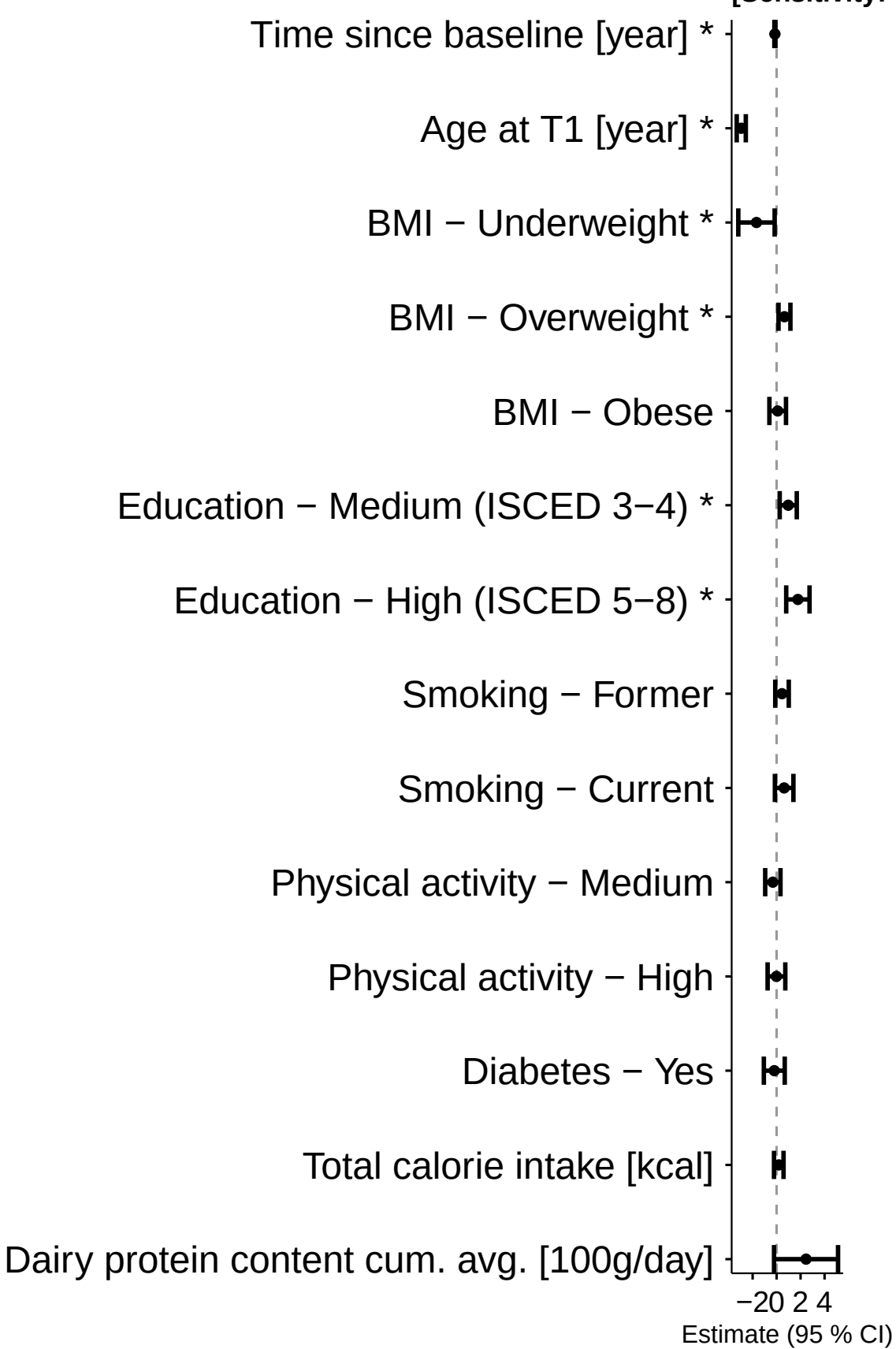
Formula – [Sensitivity: dairy <= 1000 g/day] HGS_MAX ~ prot_content_dairy_100g (linear) | main

HGS_MAX ~ prot_content_dairy_100g + time_since_baseline + age_at_baseline_scaled +
BMI_category + education_level + smoking_status + pa_levels_tertile_f1 +
diabetes_status + sumtot1_scaled + (1 | pt)

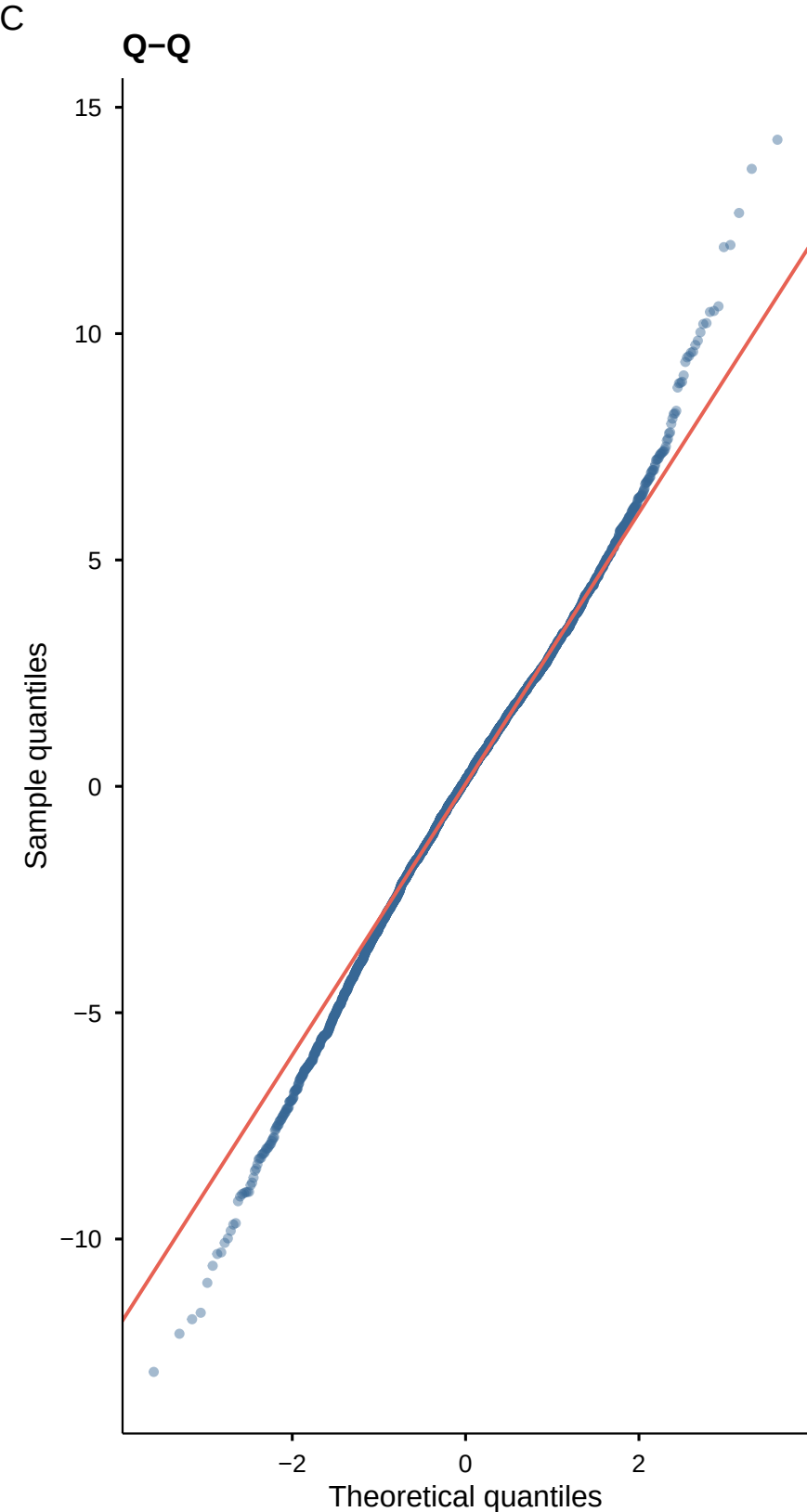
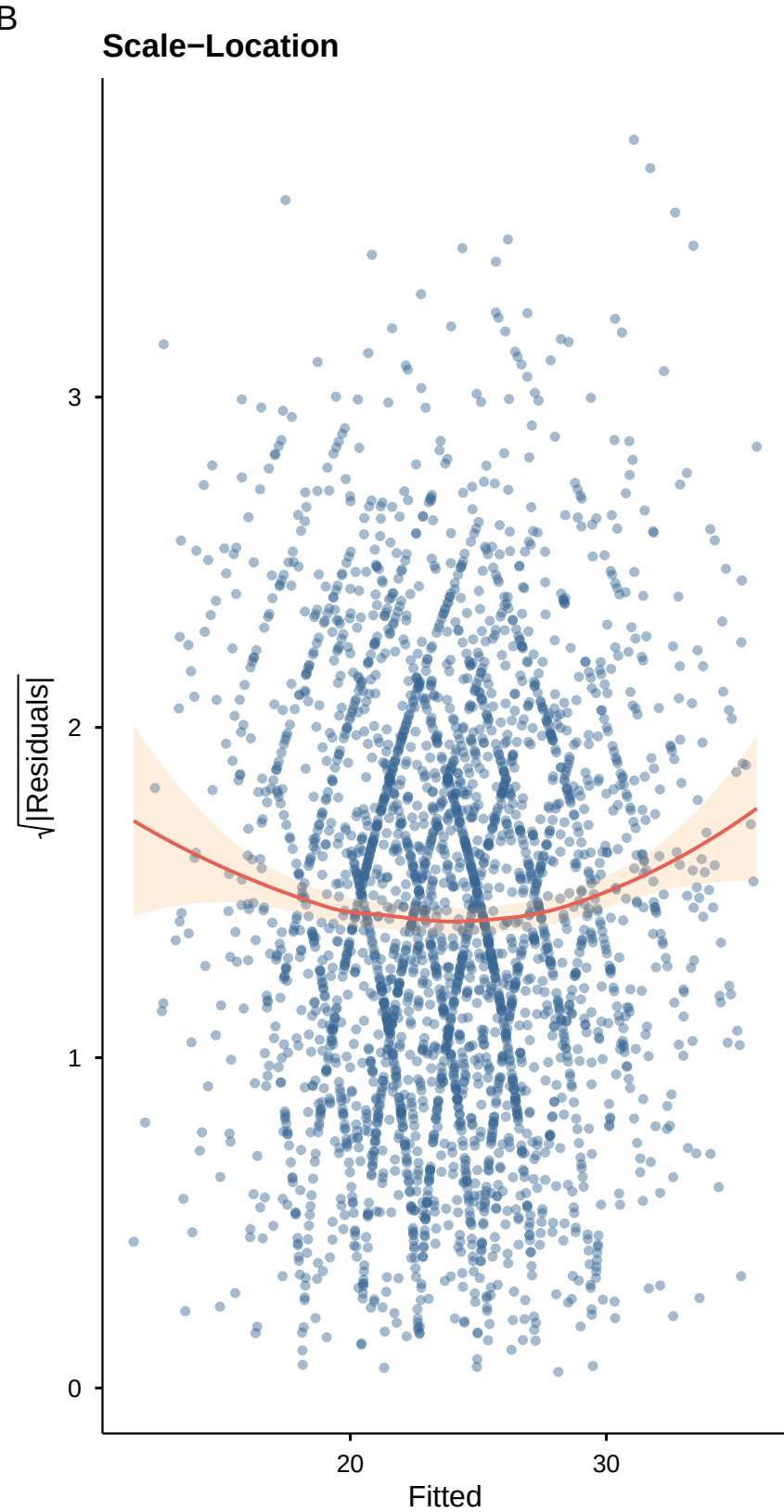
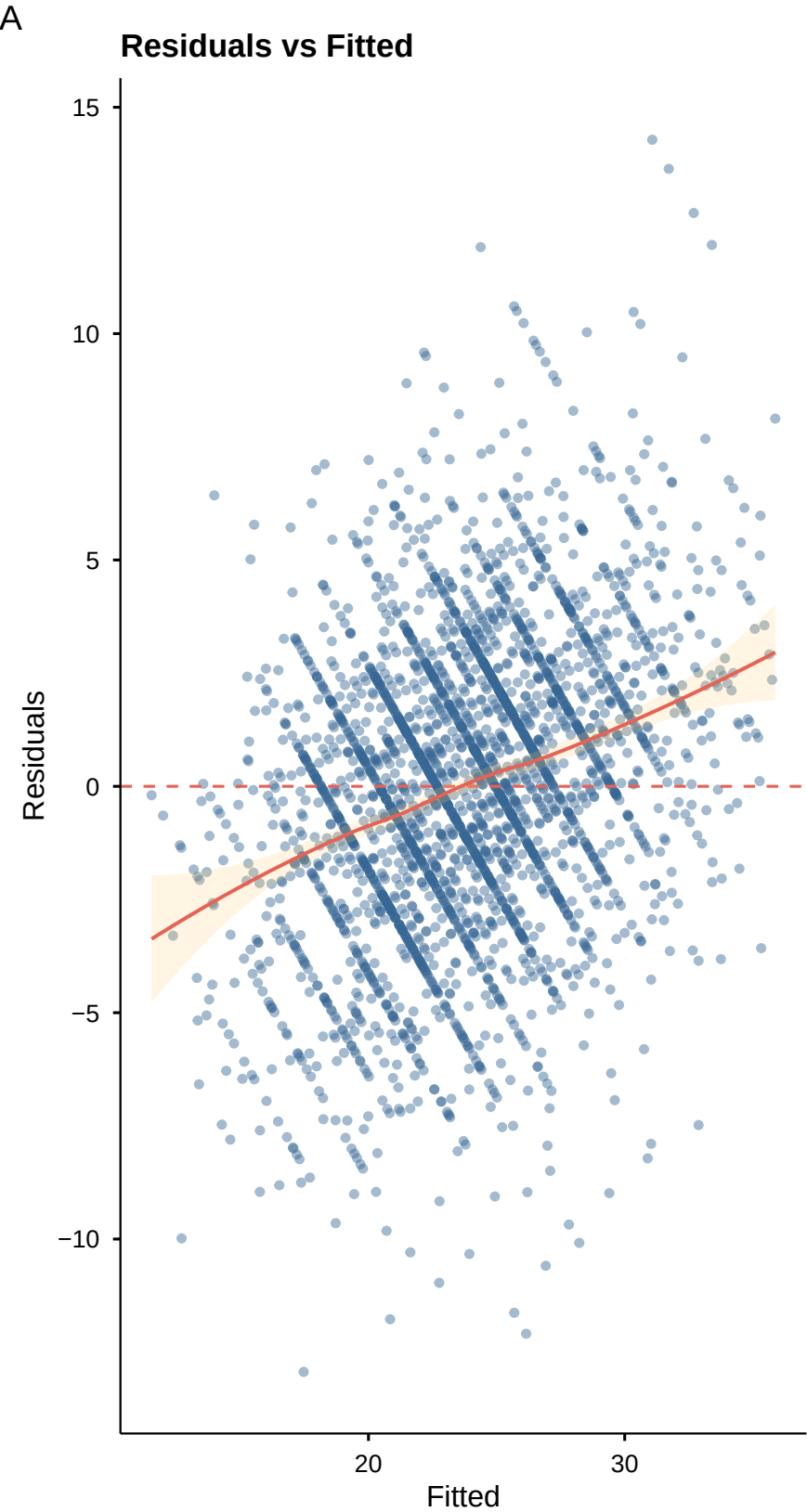
Imputations pooled: 10

Participants in sensitivity sample: 1130

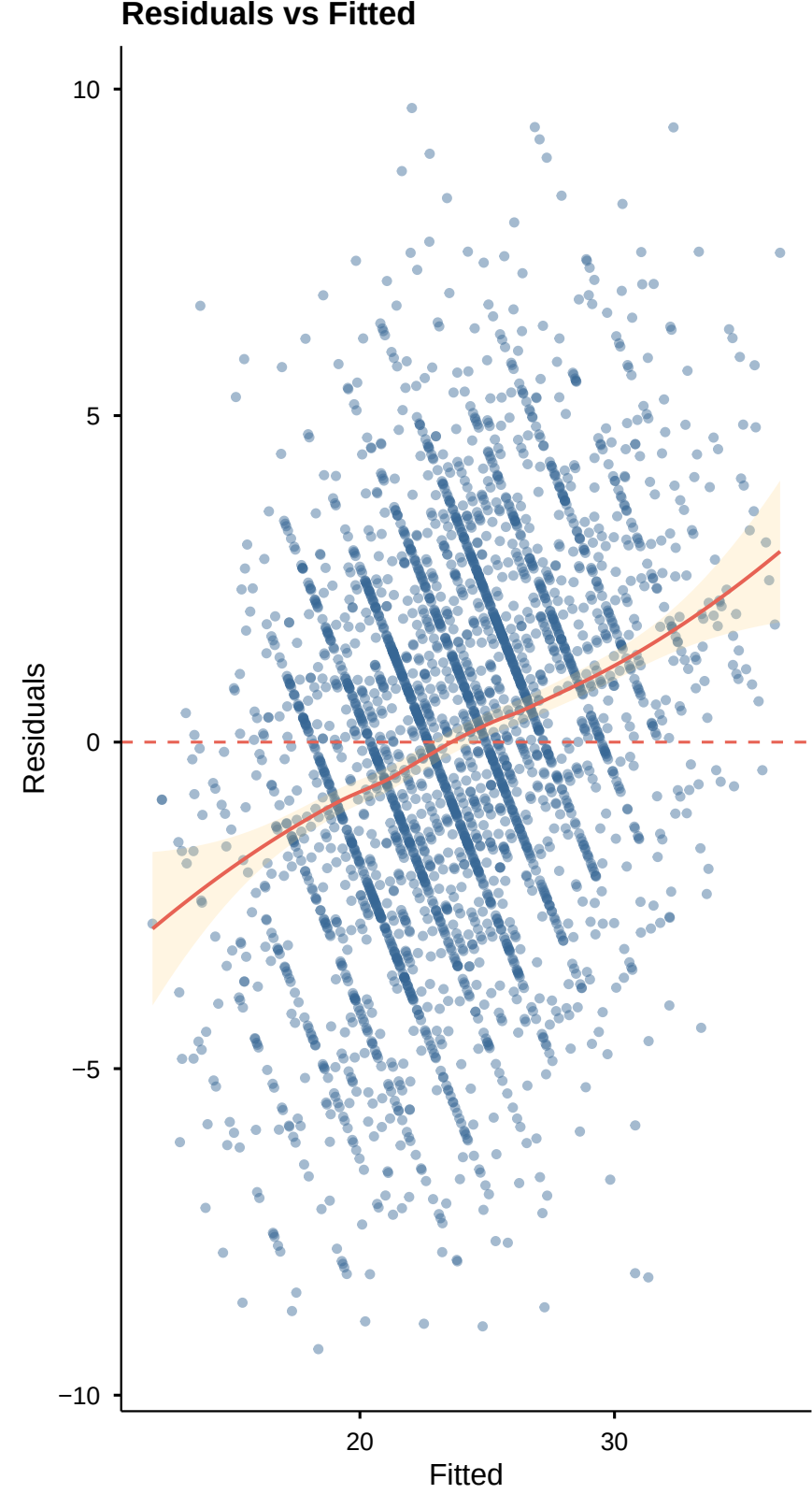
Threshold: dairy_total_gday_cumavg <= 1000 g/day



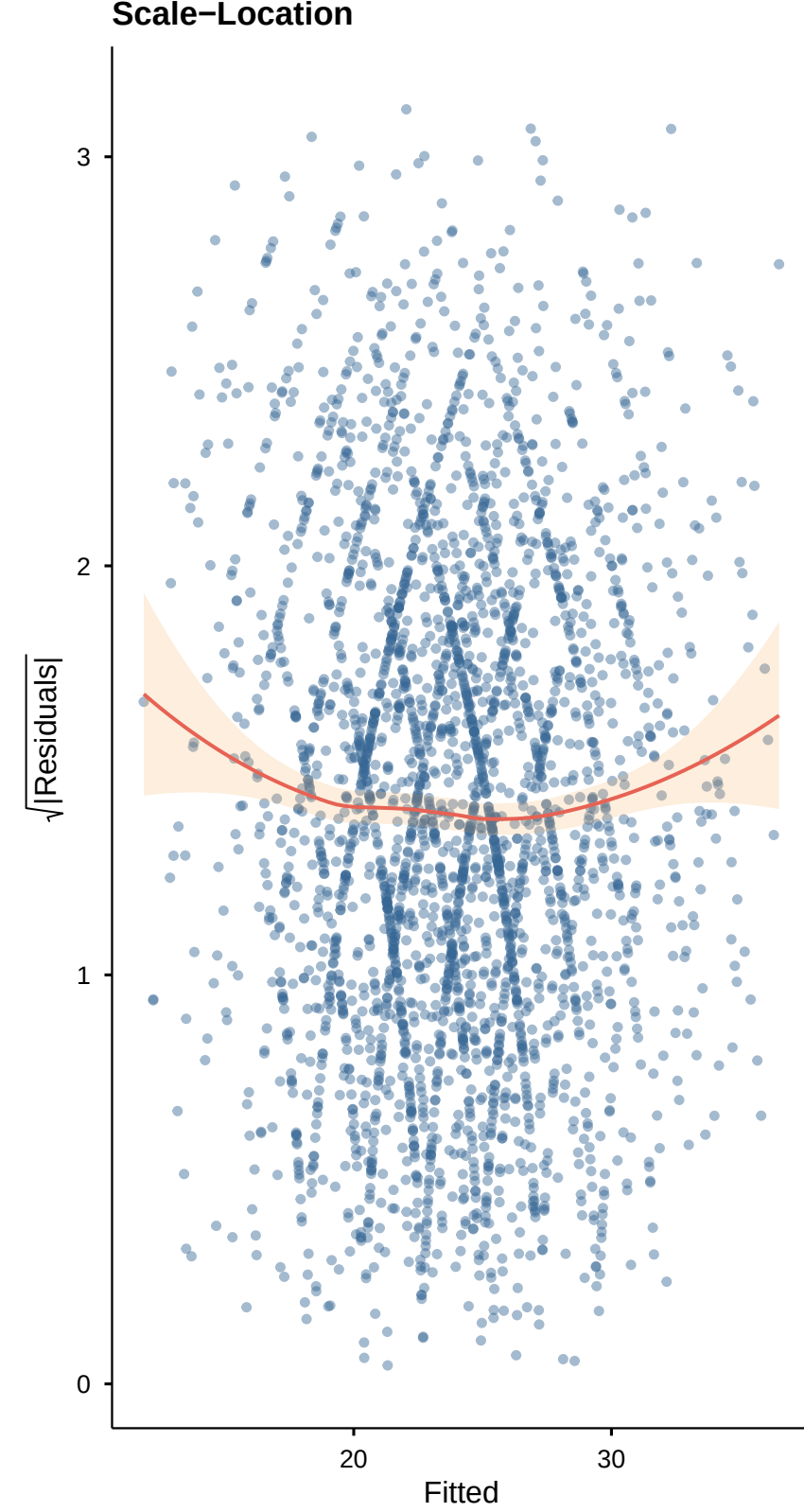
term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	23.036	0.458	50.319	1847.3	<0.001	22.139	23.934
prot_content_dairy_100g	2.456	1.360	1.806	882.7	0.071	-0.213	5.125
time_since_baseline	-0.140	0.023	-6.151	1123.7	<0.001	-0.185	-0.095
age_at_baseline_scaled	-2.949	0.193	-15.272	1549.8	<0.001	-3.327	-2.570
BMI_categoryUnderweight	-1.679	0.773	-2.172	2092.1	0.03	-3.195	-0.163
BMI_categoryOverweight	0.656	0.253	2.594	815.1	0.01	0.160	1.152
BMI_categoryObese	0.091	0.353	0.258	1946.3	0.797	-0.602	0.784
education_levelMedium (ISCED 3–4)	0.971	0.363	2.670	2692.5	0.008	0.258	1.683
education_levelHigh (ISCED 5–8)	1.774	0.496	3.576	2153.0	<0.001	0.801	2.748
smoking_statusFormer	0.451	0.288	1.564	2020.0	0.118	-0.114	1.016
smoking_statusCurrent	0.630	0.394	1.601	875.6	0.11	-0.143	1.403
pa_levels_tertile_f1Medium	-0.315	0.325	-0.969	2302.4	0.333	-0.952	0.322
pa_levels_tertile_f1High	-0.012	0.375	-0.032	311.6	0.974	-0.750	0.726
diabetes_statusDiabetes	-0.193	0.444	-0.433	2387.9	0.665	-1.064	0.679
sumtot1_scaled	0.181	0.202	0.899	245.2	0.37	-0.216	0.579



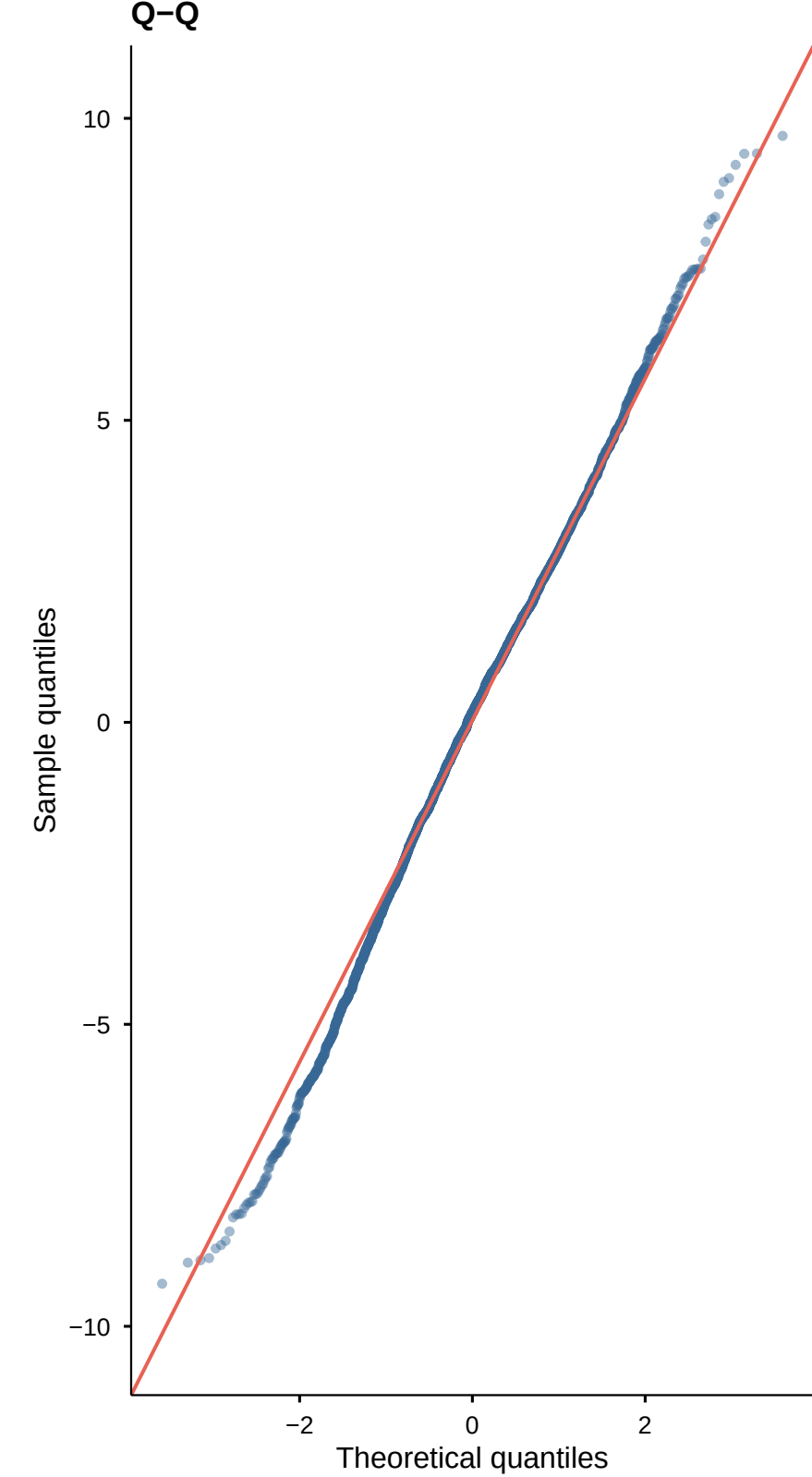
A

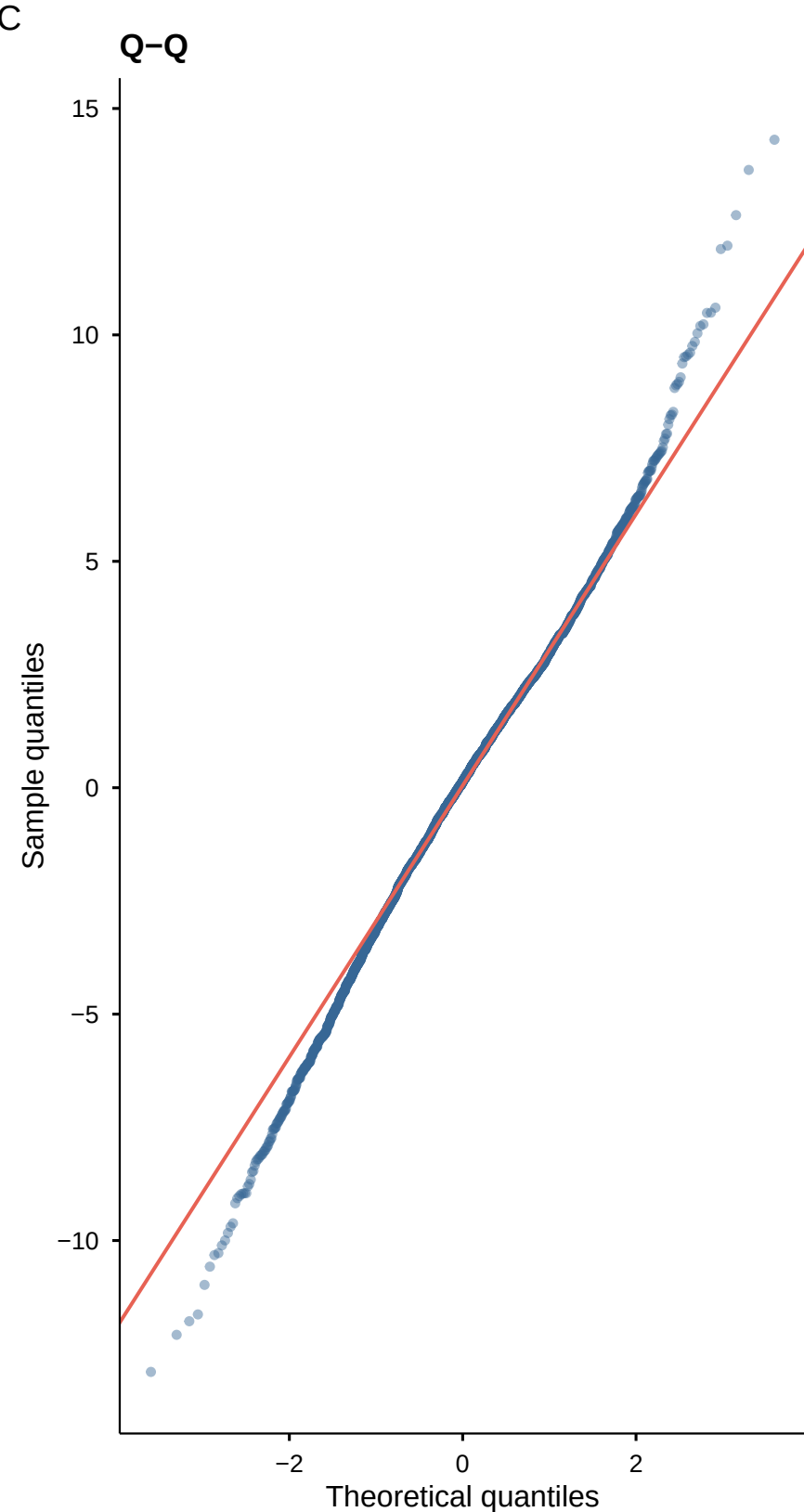
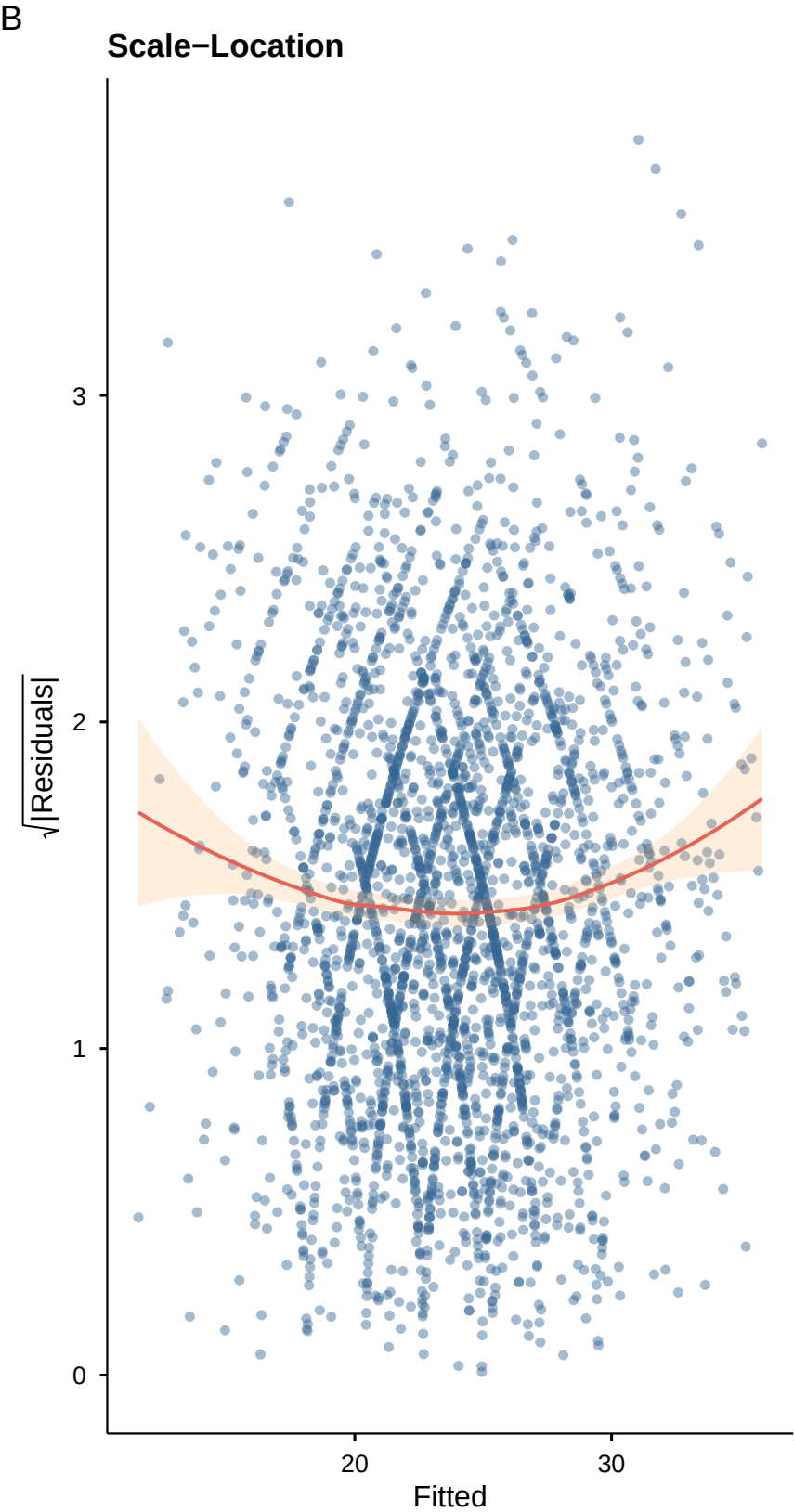
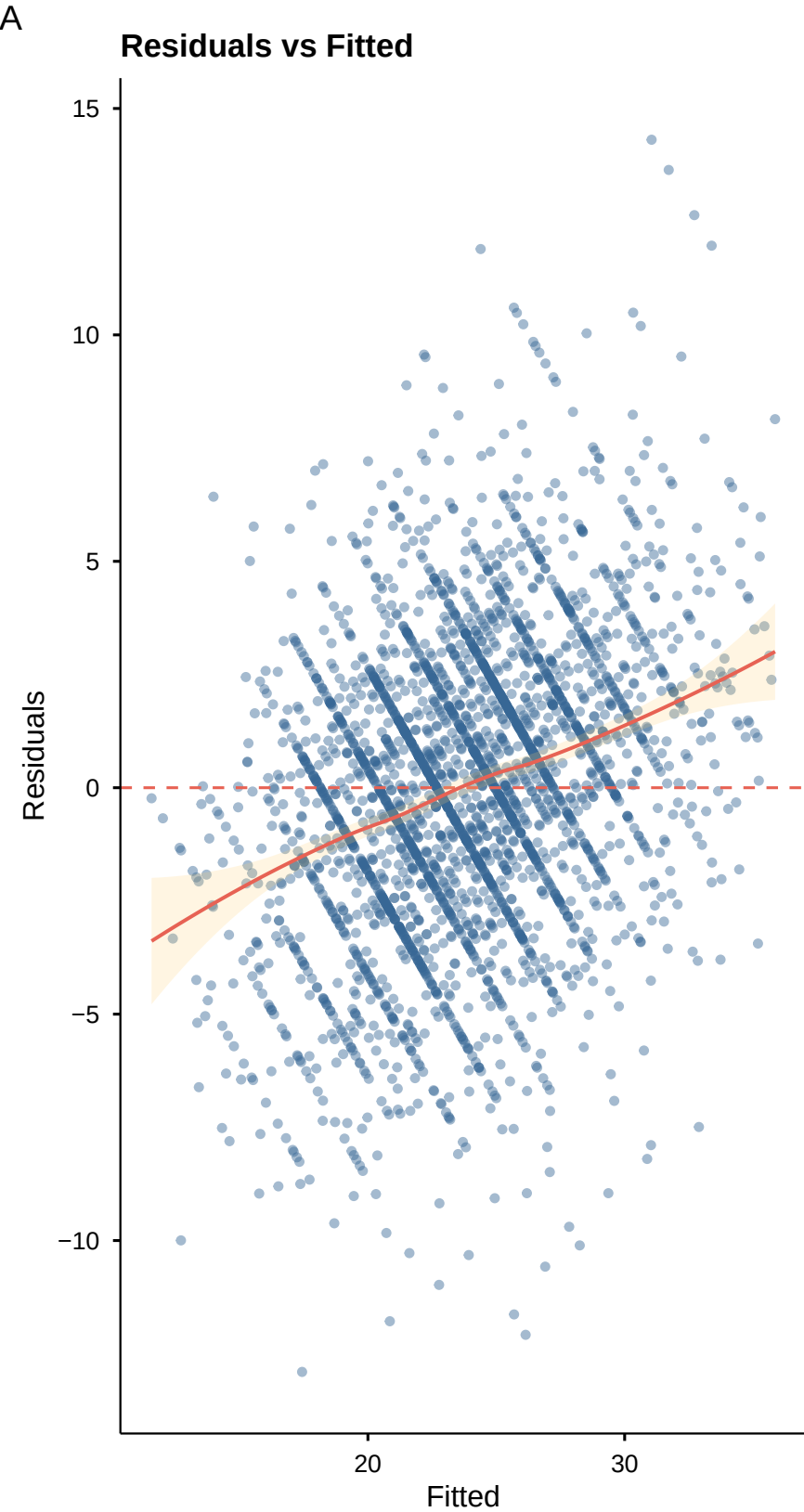


B



C





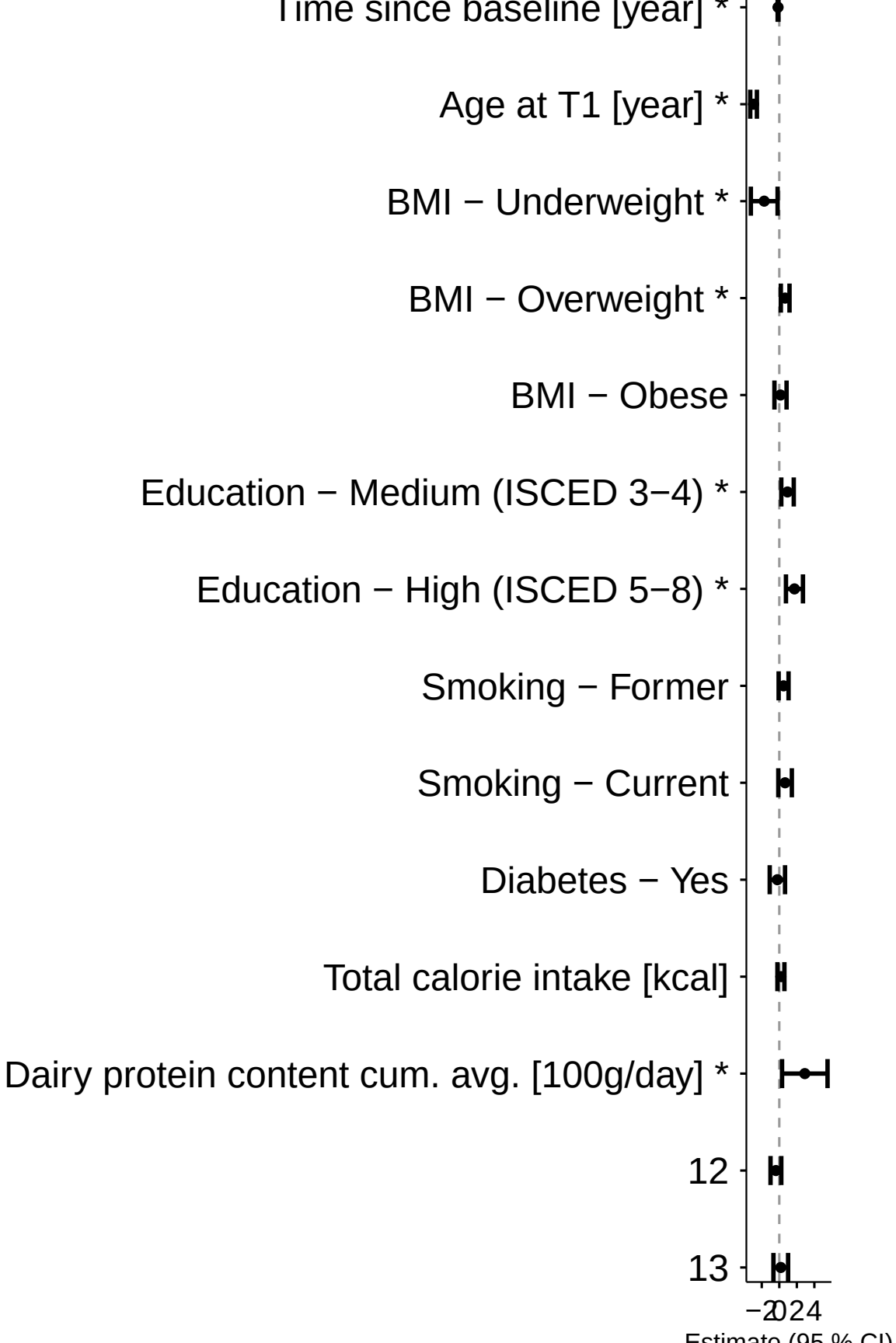
Covariate set: other_PA

age_at_baseline_scaled · BMI_category · education_level · smoking_status · pa_levels_who_f1 · diabetes_status · sumtot1_scaled

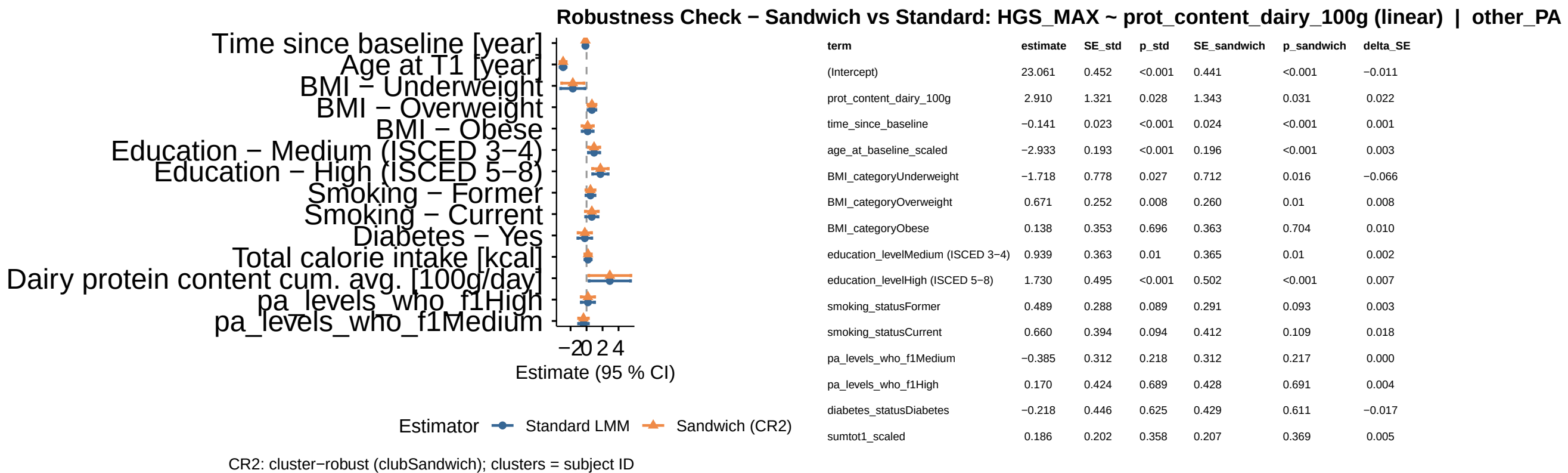
Formula – HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA

HGS_MAX ~ prot_content_dairy_100g + time_since_baseline + age_at_baseline_scaled +
BMI_category + education_level + smoking_status + pa_levels_who_f1 +
diabetes_status + sumtot1_scaled + (1 | pt)

Imputations pooled: 10



term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	23.061	0.452	51.016	1946.4	<0.001	22.175	23.948
prot_content_dairy_100g	2.910	1.321	2.202	806.8	0.028	0.316	5.503
time_since_baseline	-0.141	0.023	-6.187	1147.4	<0.001	-0.185	-0.096
age_at_baseline_scaled	-2.933	0.193	-15.164	1422.1	<0.001	-3.312	-2.554
BMI_categoryUnderweight	-1.718	0.778	-2.209	1758.4	0.027	-3.243	-0.193
BMI_categoryOverweight	0.671	0.252	2.664	870.9	0.008	0.177	1.165
BMI_categoryObese	0.138	0.353	0.391	1886.1	0.696	-0.555	0.831
education_levelMedium (ISCED 3–4)	0.939	0.363	2.587	2771.8	0.01	0.227	1.650
education_levelHigh (ISCED 5–8)	1.730	0.495	3.497	2255.9	<0.001	0.760	2.700
smoking_statusFormer	0.489	0.288	1.700	2013.2	0.089	-0.075	1.054
smoking_statusCurrent	0.660	0.394	1.676	835.0	0.094	-0.113	1.433
pa_levels_who_f1Medium	-0.385	0.312	-1.234	914.6	0.218	-0.998	0.228
pa_levels_who_f1High	0.170	0.424	0.401	284.8	0.689	-0.664	1.005
diabetes_statusDiabetes	-0.218	0.446	-0.489	2219.8	0.625	-1.092	0.656
sumtot1_scaled	0.186	0.202	0.921	221.9	0.358	-0.212	0.585



Variance Components – HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA

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.. Full VarCorr output .....
Groups   Name          Variance Std.Dev.
pt       (Intercept) 14.776    3.8439
Residual                14.342    3.7871

.. Summary .....
Random intercept variance: 14.77591 (SD = 3.84394)
Random slope: not included in this model
Residual variance:          14.34198 (SD = 3.78708)
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Effect Size (R²) – HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA

Marginal R² (fixed effects only): 0.1623

Conditional R² (fixed + random effects): 0.5874

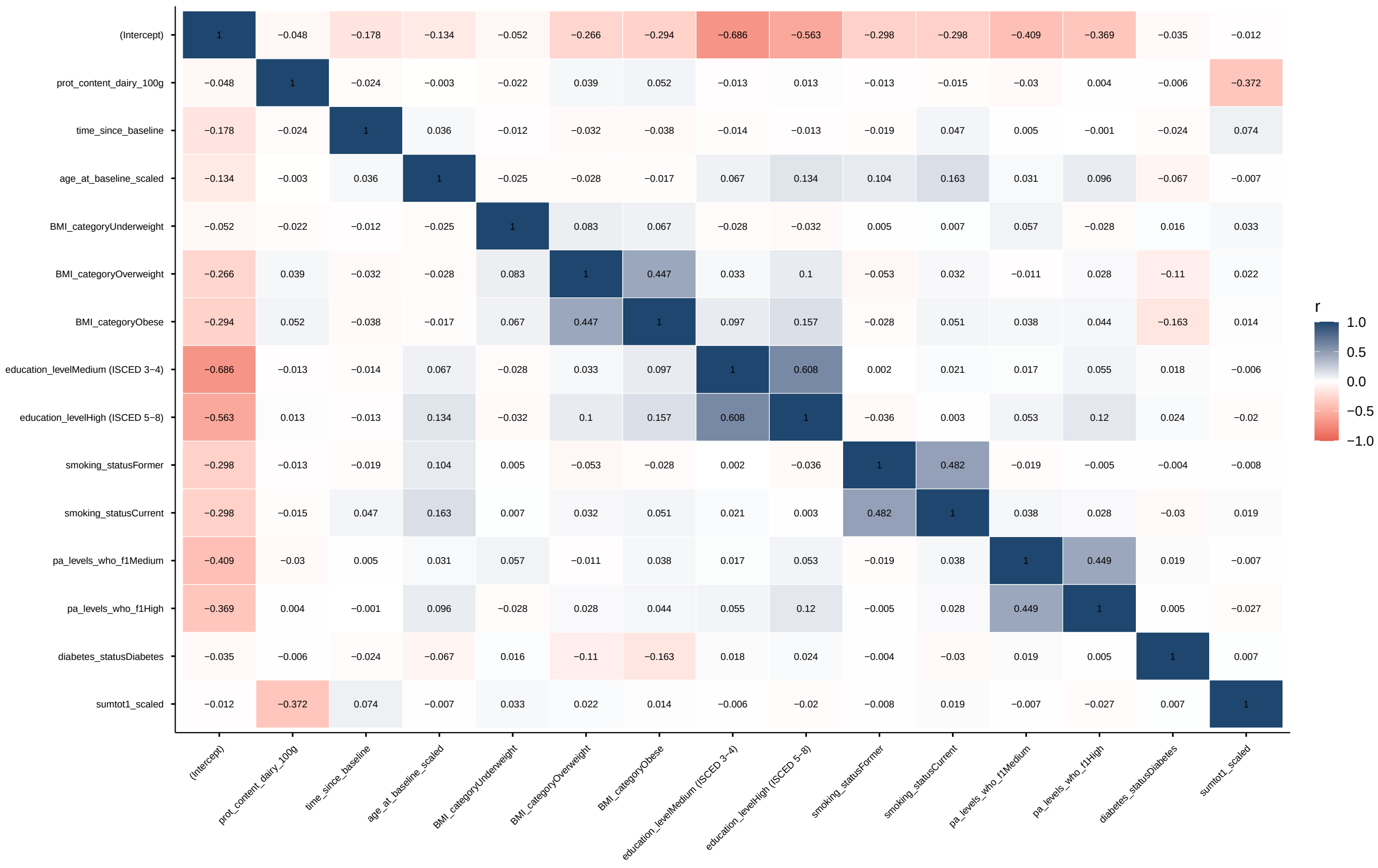
Method: Nakagawa & Schielzeth (2013), Johnson (2014)

VIF – HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA

VIF > 5 flagged with '!'

	term	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$	adjusted_vif	flag
	prot_content_dairy_100g	1.169084	1	1.081242	1.081	
	time_since_baseline	1.014459	1	1.007203	1.007	
	age_at_baseline_scaled	1.066009	1	1.032477	1.032	
	BMI_category	1.097885	3	1.015686	1.016	
	education_level	1.071187	2	1.017341	1.017	
	smoking_status	1.050677	2	1.012435	1.012	
	pa_levels_who_f1	1.037834	2	1.009327	1.009	
	diabetes_status	1.041412	1	1.020496	1.020	
	sumtot1_scaled	1.172161	1	1.082664	1.083	

Fixed-Effect Correlations – HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA



Influential Points – HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA

Influential observations (top 1% Cook's D > 0.19965): 32

Index Cook's D

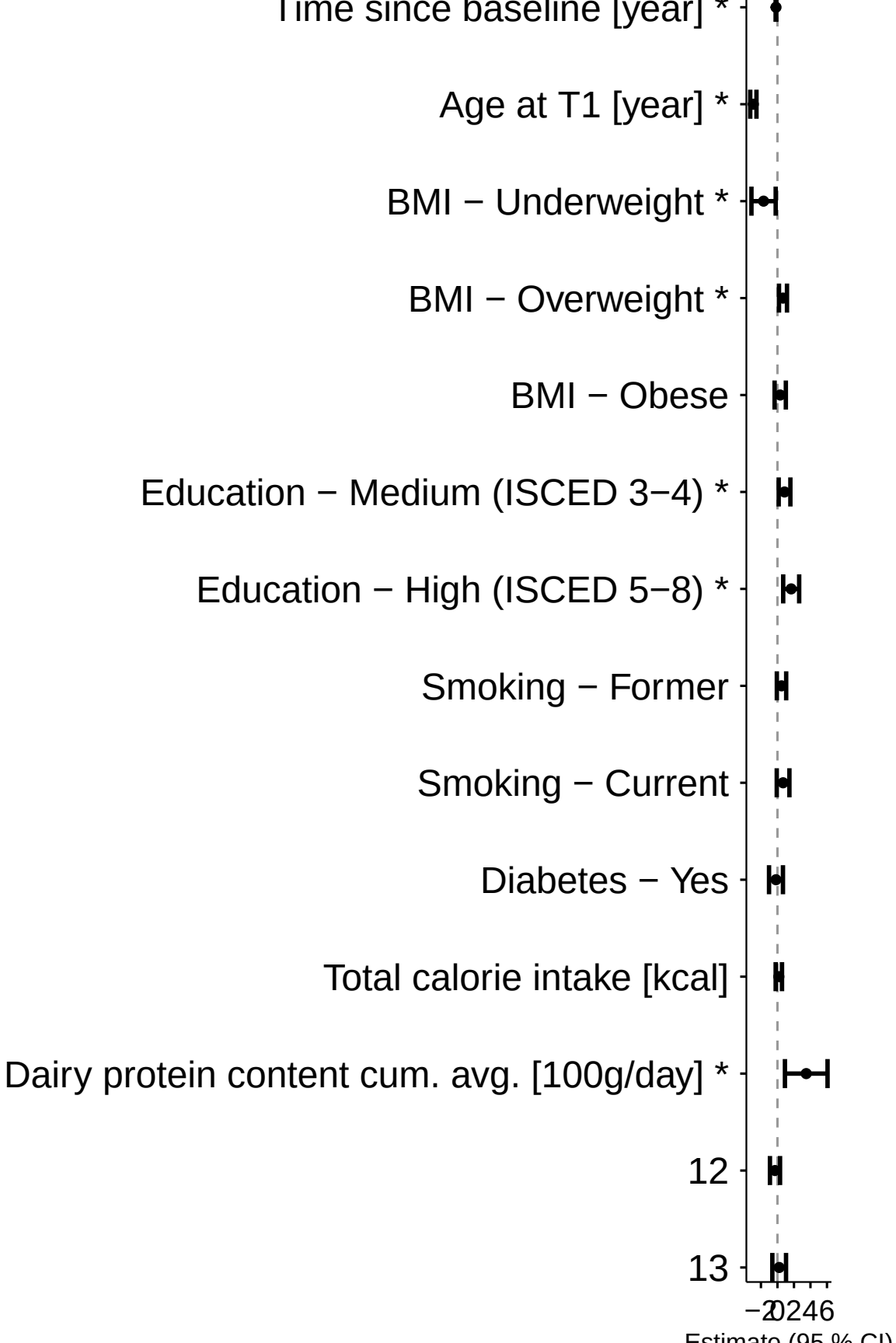
index	cooks_d
22	0.28801
516	0.25400
837	0.36083
896	0.29292
1227	0.29334
1252	0.43872
1428	0.21040
1758	0.23862
1782	0.20704
1900	0.22771
1978	0.30998
1993	0.22068
2051	0.22182
2076	0.20439
2091	0.21032
2100	0.30328
2147	0.20009
2184	0.23102
2297	0.33183
2319	0.23334
2323	0.21406
2334	0.27819
2353	0.38911
2399	0.34142
2457	0.21641
2465	0.20491
2570	0.21114
2692	0.34554
2752	0.23955
2753	0.20371
2759	0.30236
2785	0.26161

Formula – [Excl. influential (top 1% Cook's D)] HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA

Excluded participants (top 1% Cook's D, imp 1): 31

Remaining n: 1104

HGS_MAX ~ prot_content_dairy_100g + time_since_baseline + age_at_baseline_scaled +
BMI_category + education_level + smoking_status + pa_levels_who_f1 +
diabetes_status + sumtot1_scaled + (1 | pt)



term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	23.167	0.448	51.677	1939.7	<0.001	22.288	24.046
prot_content_dairy_100g	3.487	1.314	2.653	689.5	0.008	0.906	6.068
time_since_baseline	-0.178	0.022	-8.267	1207.0	<0.001	-0.220	-0.136
age_at_baseline_scaled	-2.919	0.192	-15.187	1734.7	<0.001	-3.296	-2.542
BMI_categoryUnderweight	-1.681	0.748	-2.247	1584.4	0.025	-3.149	-0.214
BMI_categoryOverweight	0.671	0.247	2.718	792.3	0.007	0.186	1.155
BMI_categoryObese	0.335	0.347	0.965	1712.2	0.334	-0.346	1.016
education_levelMedium (ISCED 3–4)	0.859	0.361	2.380	2665.9	0.017	0.151	1.567
education_levelHigh (ISCED 5–8)	1.664	0.494	3.371	2343.3	<0.001	0.696	2.632
smoking_statusFormer	0.491	0.288	1.704	1867.1	0.089	-0.074	1.057
smoking_statusCurrent	0.683	0.391	1.748	782.5	0.081	-0.084	1.450
pa_levels_who_f1Medium	-0.296	0.312	-0.951	896.1	0.342	-0.908	0.315
pa_levels_who_f1High	0.211	0.422	0.500	323.4	0.617	-0.619	1.041
diabetes_statusDiabetes	-0.174	0.430	-0.406	1987.3	0.685	-1.018	0.669
sumtot1_scaled	0.169	0.195	0.866	222.6	0.387	-0.215	0.553

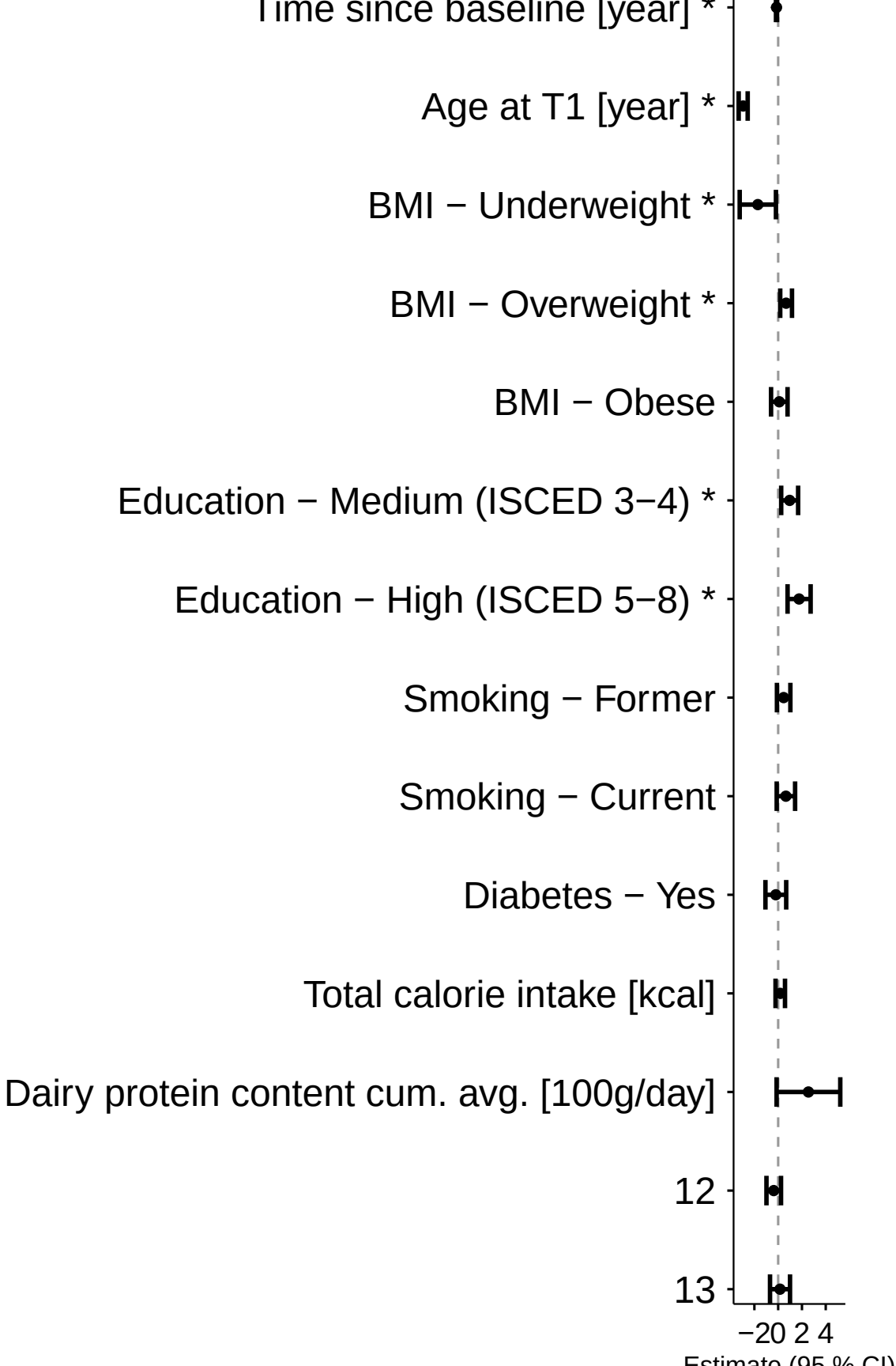
Formula – [Sensitivity: dairy <= 1000 g/day] HGS_MAX ~ prot_content_dairy_100g (linear) | other_PA

HGS_MAX ~ prot_content_dairy_100g + time_since_baseline + age_at_baseline_scaled +
BMI_category + education_level + smoking_status + pa_levels_who_f1 +
diabetes_status + sumtot1_scaled + (1 | pt)

Imputations pooled: 10

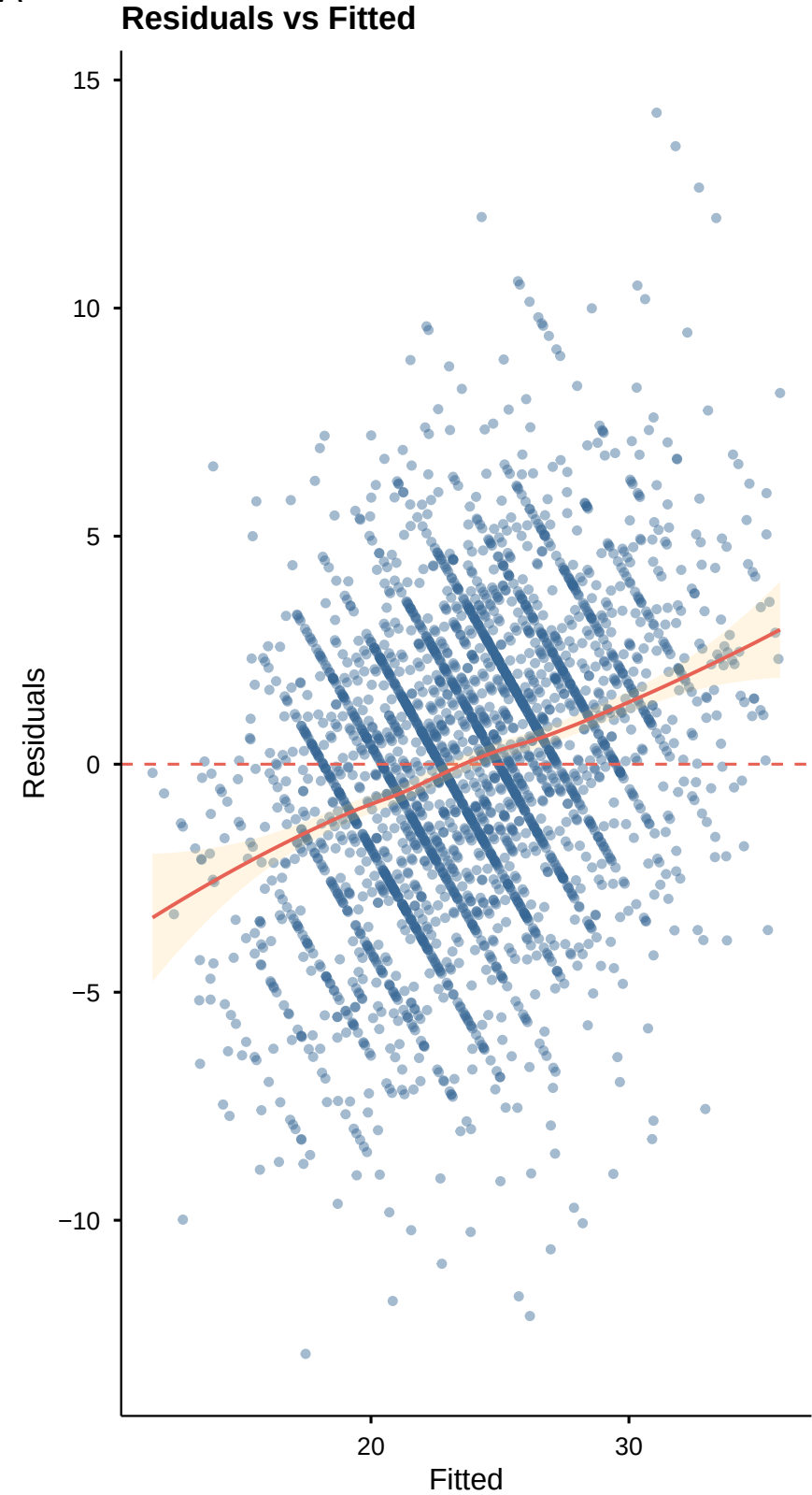
Participants in sensitivity sample: 1130

Threshold: dairy_total_gday_cumavg <= 1000 g/day

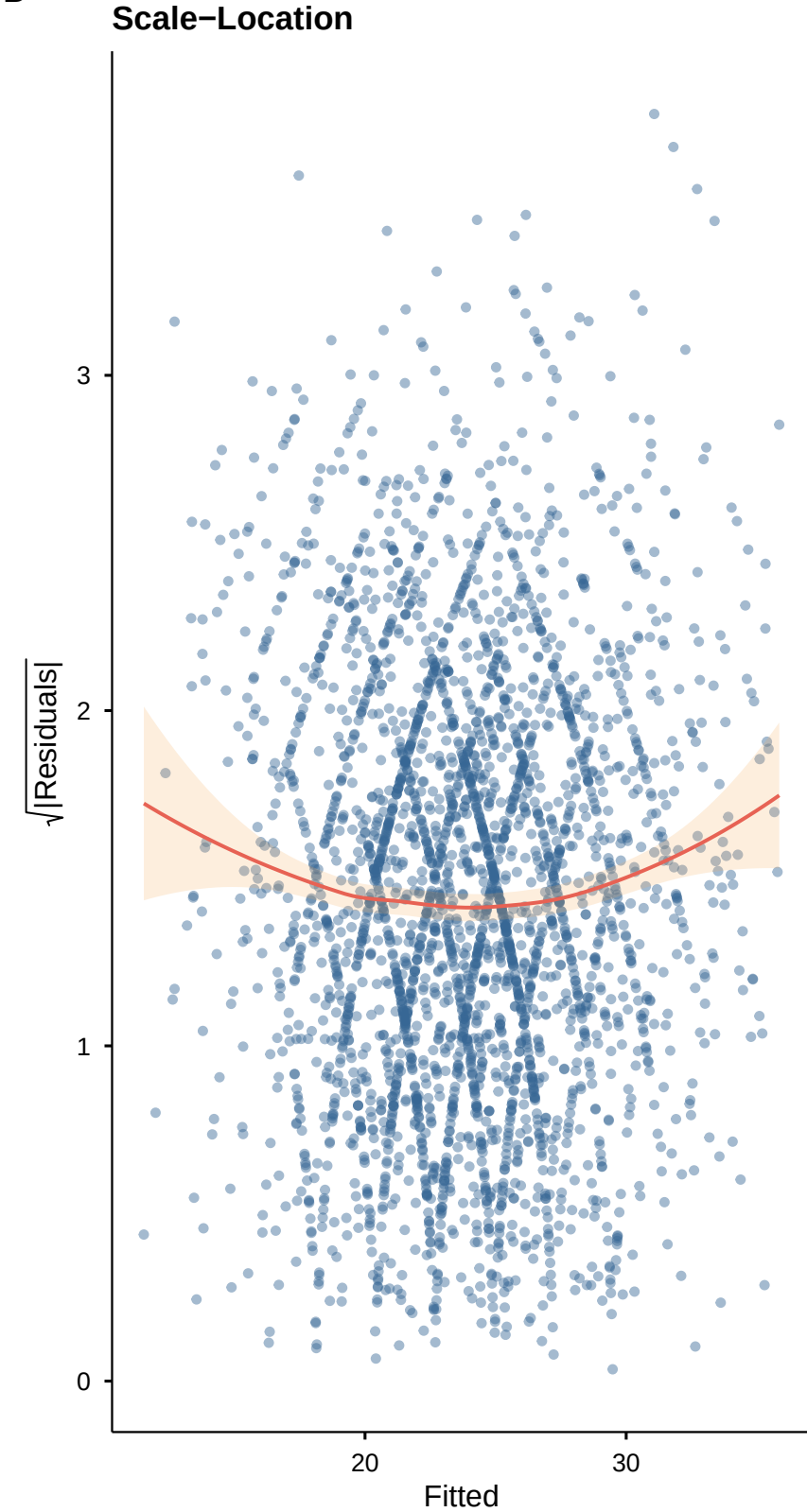


term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	23.050	0.452	50.973	1937.6	<0.001	22.163	23.937
prot_content_dairy_100g	2.545	1.364	1.866	812.4	0.062	-0.132	5.221
time_since_baseline	-0.140	0.023	-6.156	1145.2	<0.001	-0.185	-0.096
age_at_baseline_scaled	-2.944	0.193	-15.232	1412.4	<0.001	-3.323	-2.565
BMI_categoryUnderweight	-1.712	0.777	-2.202	1756.9	0.028	-3.236	-0.187
BMI_categoryOverweight	0.668	0.252	2.645	863.5	0.008	0.172	1.163
BMI_categoryObese	0.095	0.354	0.267	1891.0	0.789	-0.599	0.788
education_levelMedium (ISCED 3–4)	0.966	0.363	2.661	2728.7	0.008	0.254	1.678
education_levelHigh (ISCED 5–8)	1.760	0.494	3.559	2227.2	<0.001	0.790	2.729
smoking_statusFormer	0.460	0.288	1.597	1986.1	0.11	-0.105	1.025
smoking_statusCurrent	0.649	0.394	1.647	793.6	0.1	-0.125	1.422
pa_levels_who_f1Medium	-0.373	0.312	-1.197	954.1	0.232	-0.985	0.239
pa_levels_who_f1High	0.155	0.427	0.363	256.2	0.717	-0.685	0.995
diabetes_statusDiabetes	-0.199	0.445	-0.447	2218.7	0.655	-1.072	0.674
sumtot1_scaled	0.177	0.202	0.877	240.7	0.382	-0.221	0.575

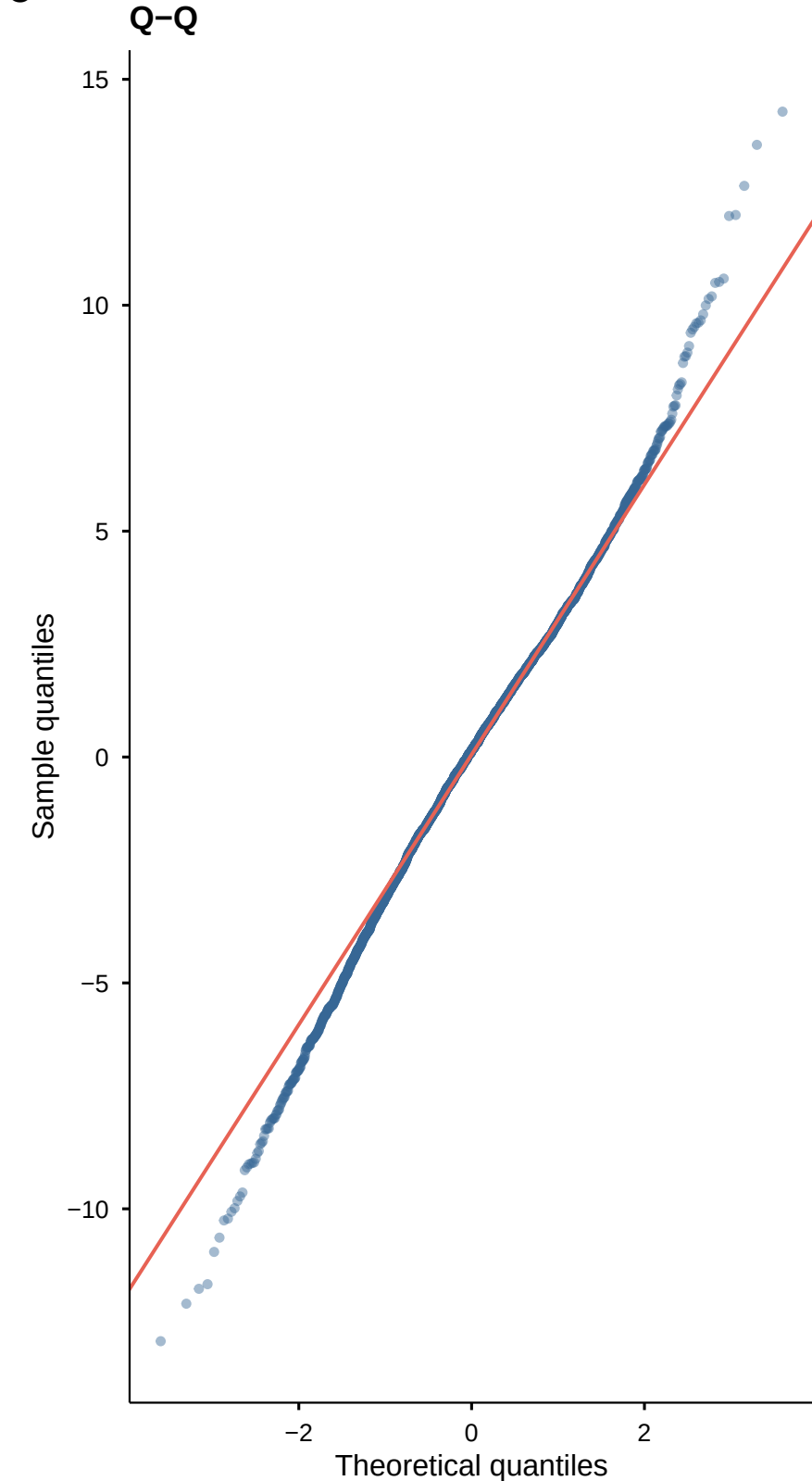
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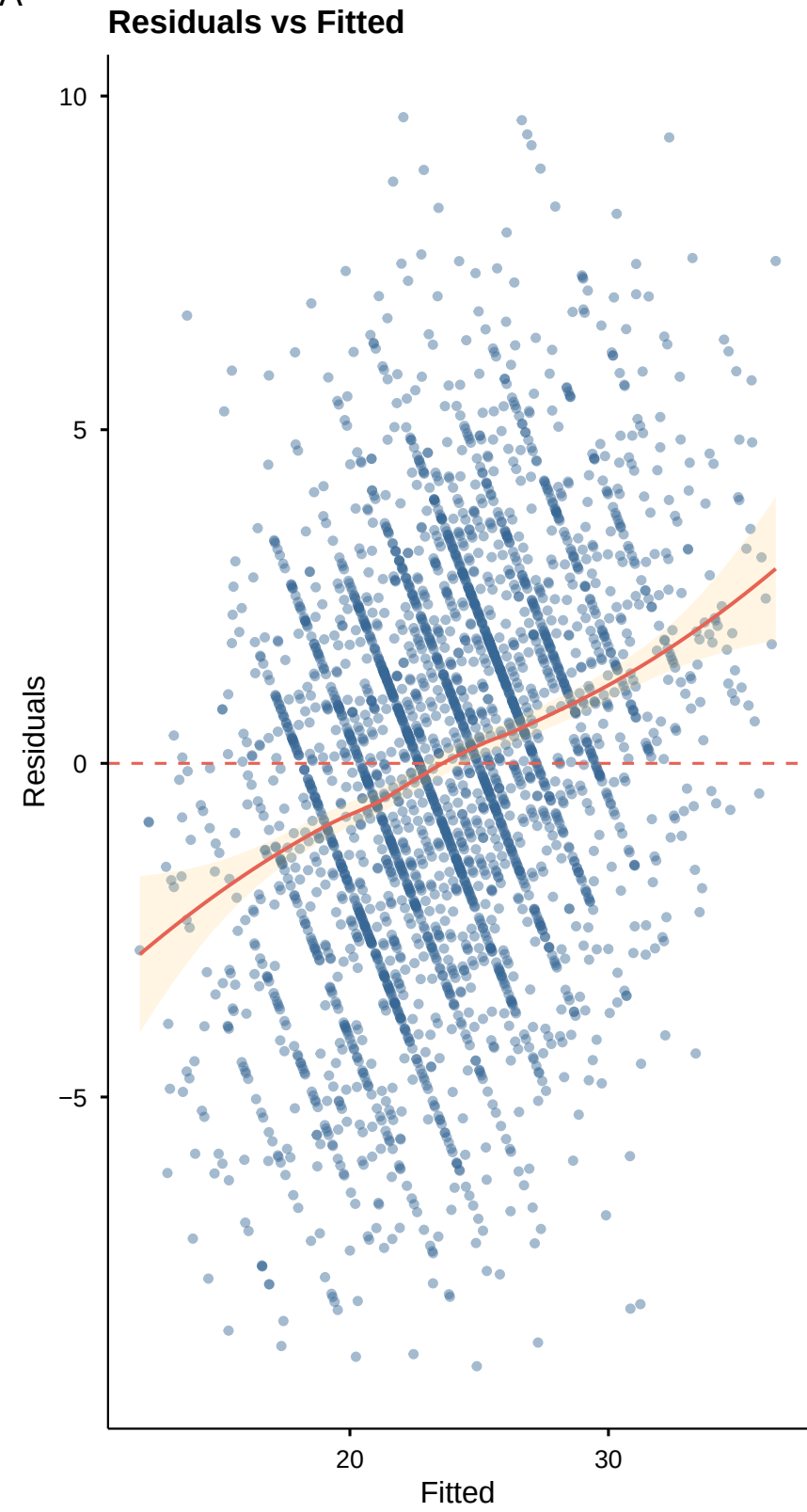
B



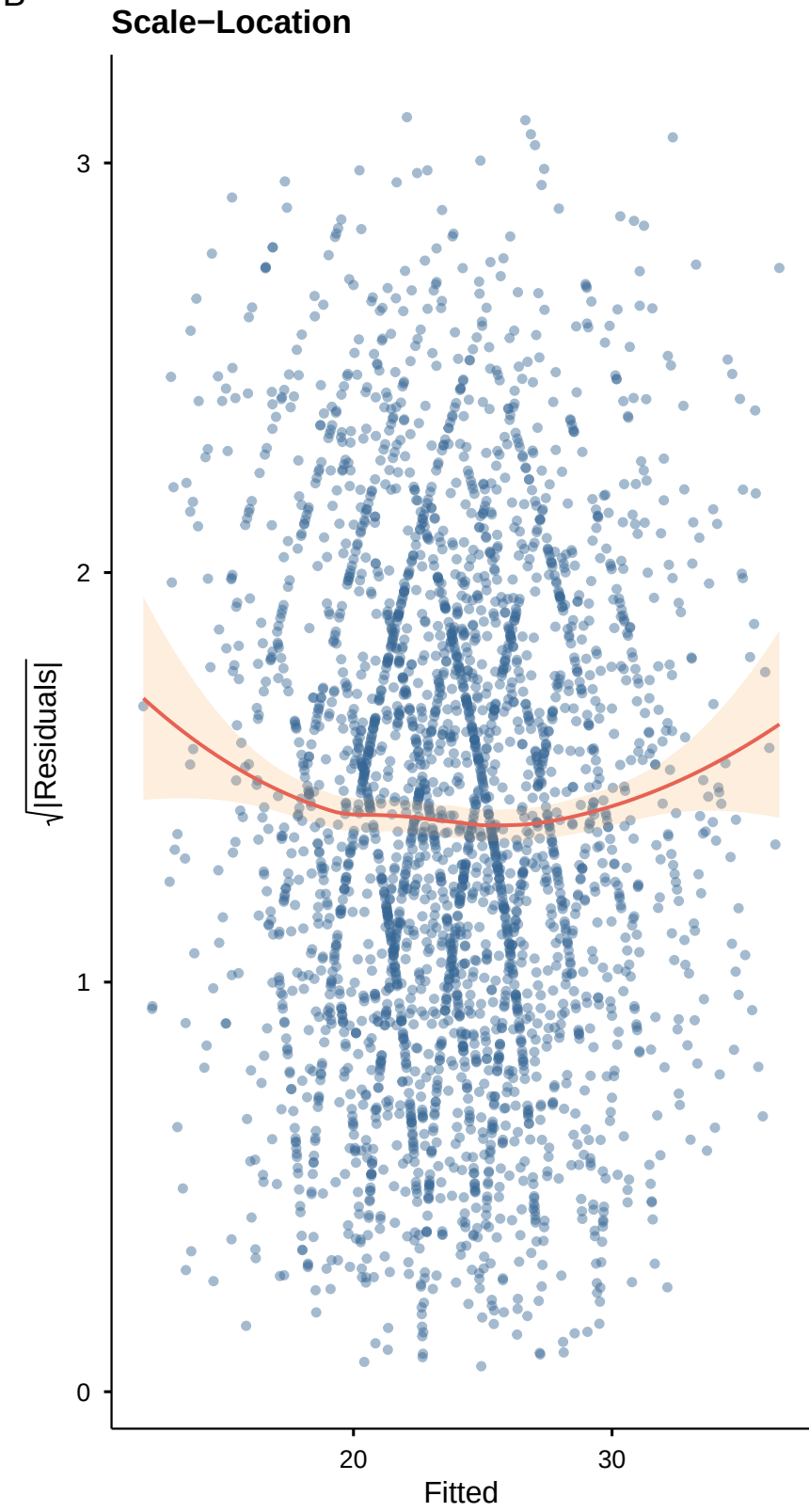
C



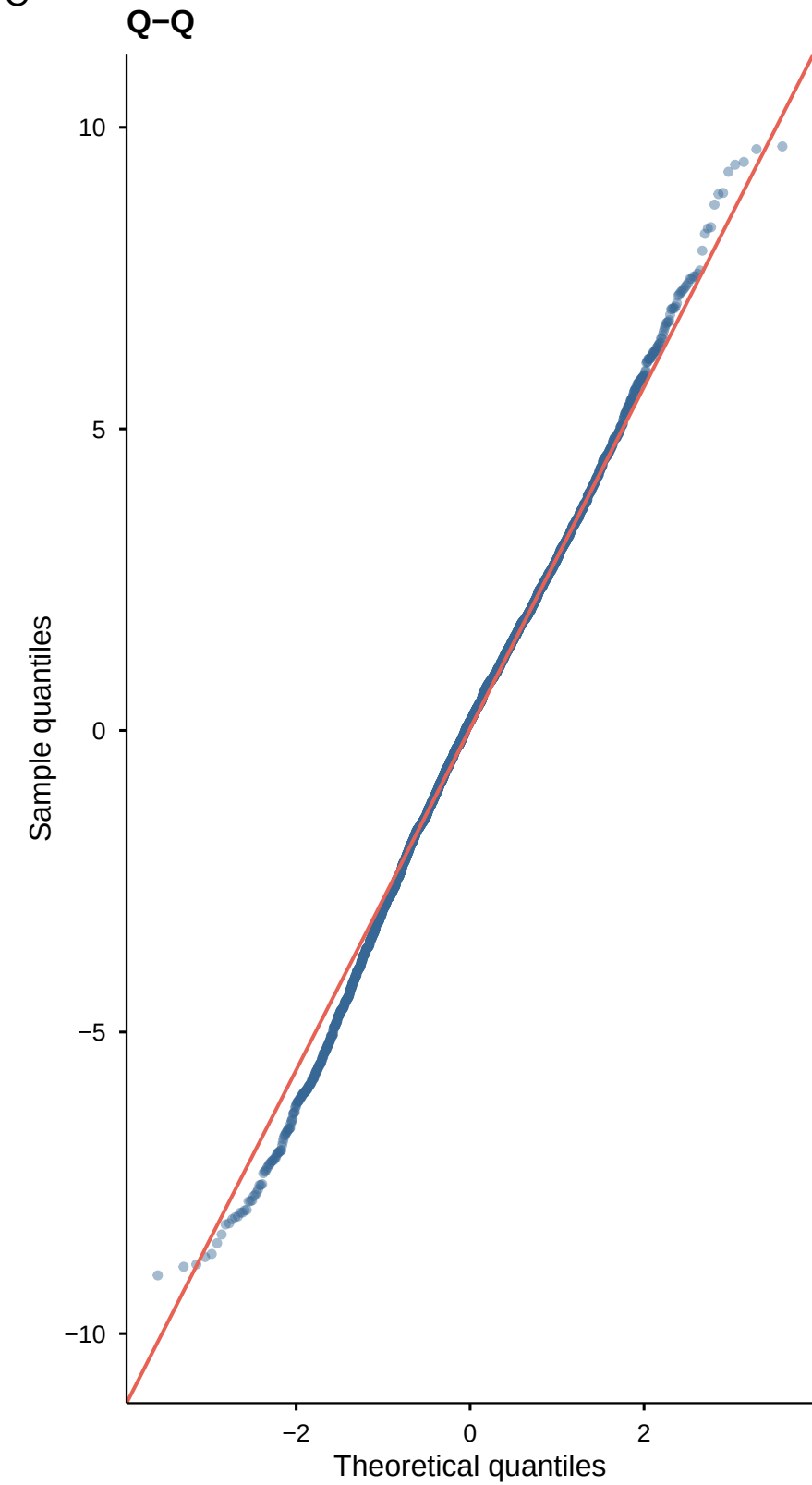
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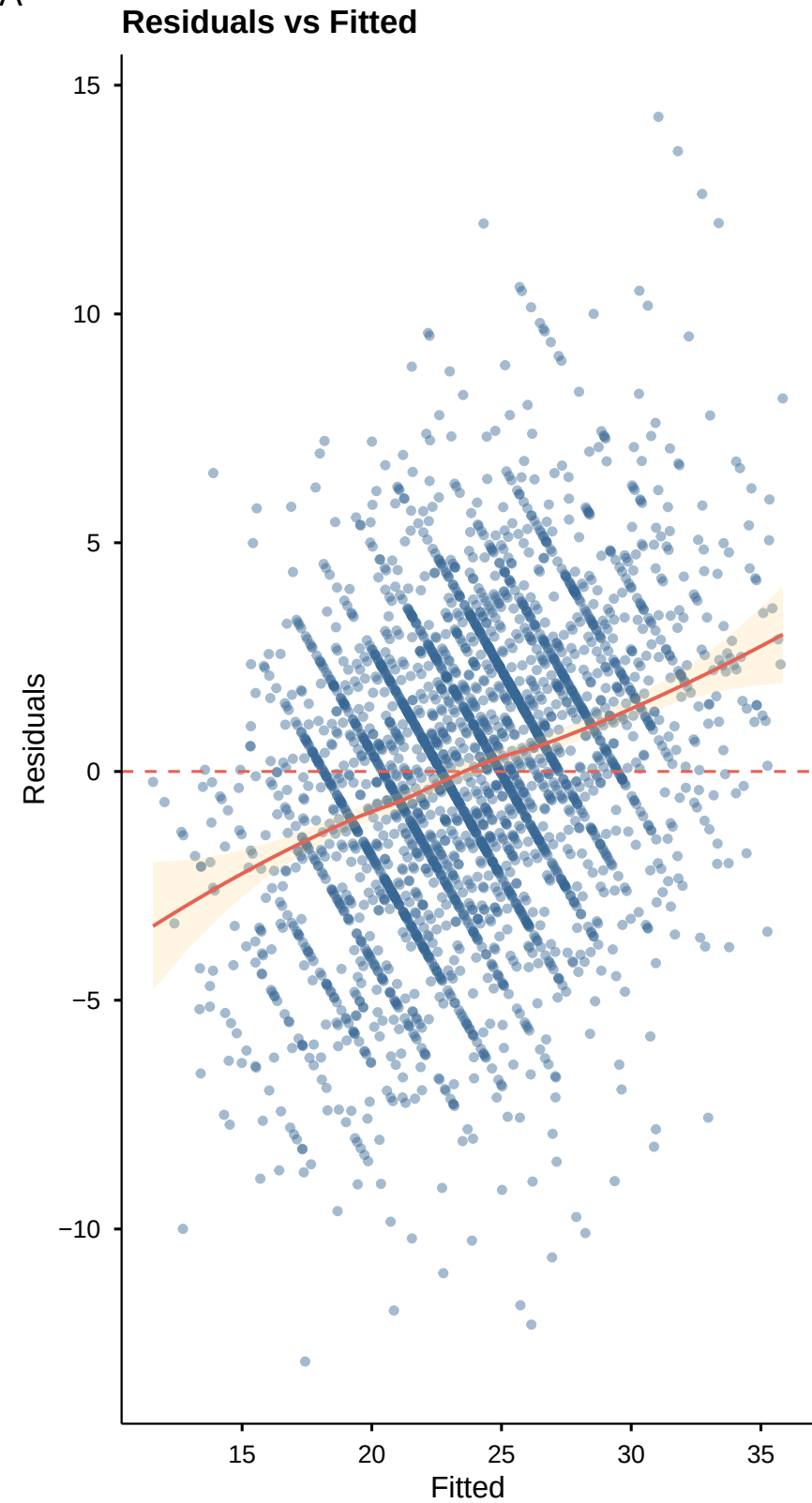
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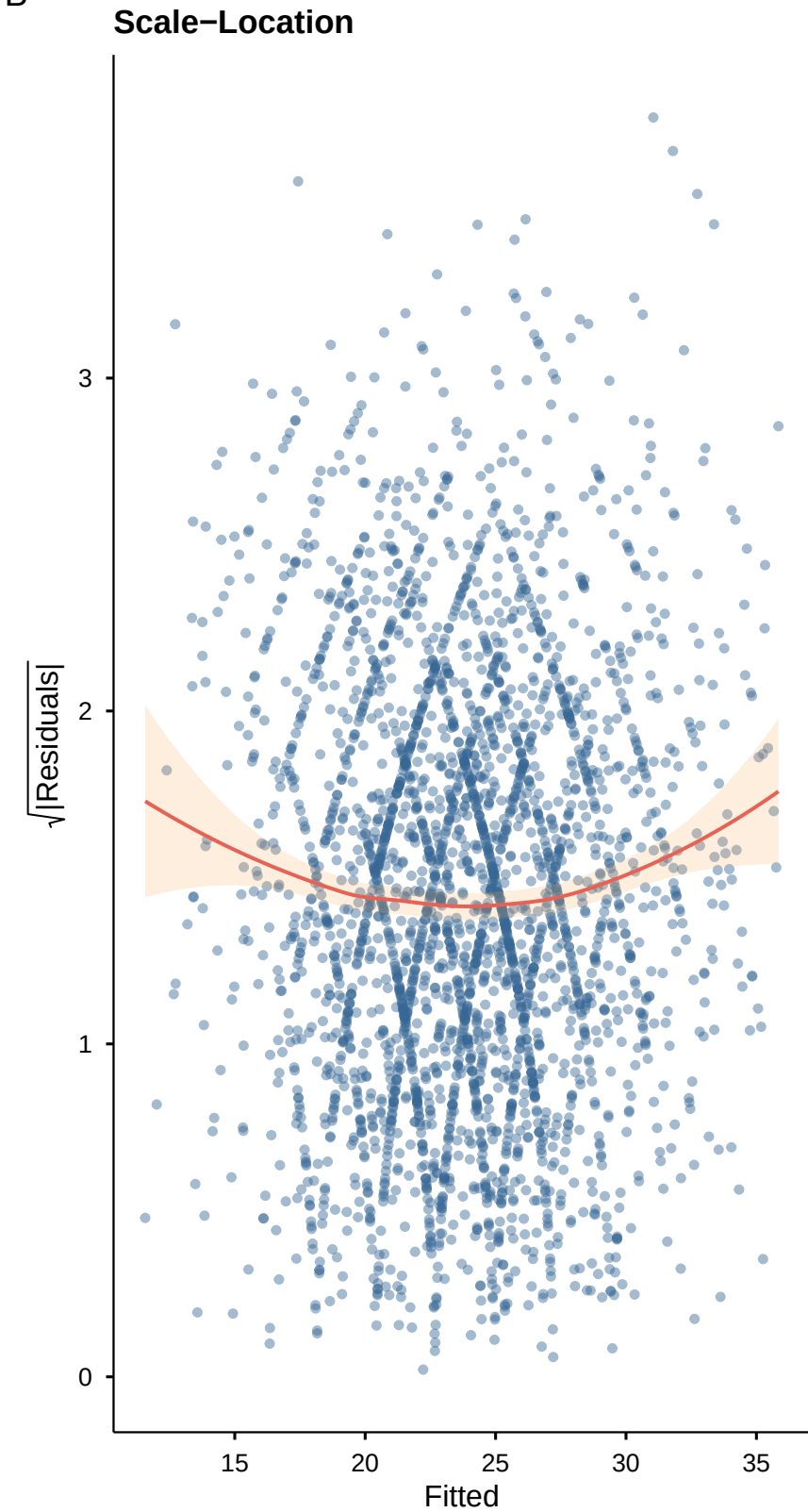
C



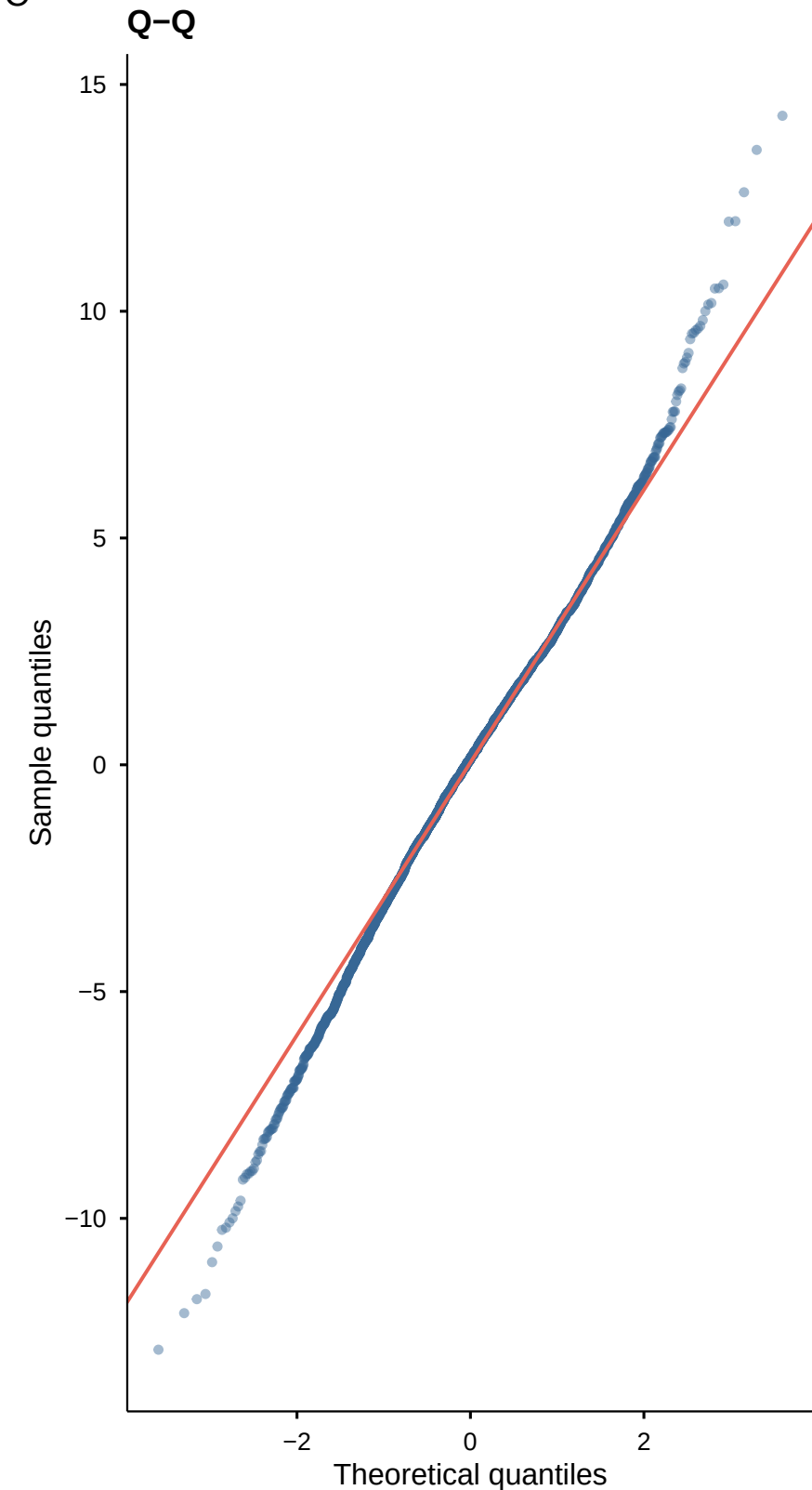
A



B



C



Session Info

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[37m [41mP [49m [39m V8	8.0.1	[90m2025-10-10 [39m [90m[?] [39m [1m [35mRSPM [39m [22m
vctrs	0.7.3	[90m2026-04-11 [39m [90m[1] [39m [1m [35mRSPM (R 4.5.0) [39m [22m
visNetwork	2.1.4	[90m2025-09-04 [39m [90m[1] [39m [1m [35mRSPM (R 4.5.0) [39m [22m
[37m [41mP [49m [39m vroom	1.7.0	[90m2026-01-27 [39m [90m[?] [39m [1m [35mRSPM [39m [22m
[37m [41mP [49m [39m WeightIt	* 1.7.0	[90m2026-04-22 [39m [90m[?] [39m [1m [35mRSPM [39m [22m
[37m [41mP [49m [39m withr	3.0.2	[90m2024-10-28 [39m [90m[?] [39m [1m [35mRSPM [39m [22m
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[90m [1] /Users/annaboss/Documents/OsteoLaus-Dairy-Sarcopenia/renv/library/macos/R-4.5/aarch64-apple-darwin20 [39m		
[90m [2] /Users/annaboss/Library/Caches/org.R-project.R/R/renv/sandbox/macos/R-4.5/aarch64-apple-darwin20/4cd76b74 [39m		

[41m [37m* [39m [49m .. Packages attached to the search path.

[41m [37mP [39m [49m .. Loaded and on-disk path mismatch.

[1m [36m..... [39m [22m